

HARI VIGNESH RAO 23BD1A052Q

WEEK-8---SNS,SQS & S3-SNS-SQS-LAMBDA SERVICES

1) Simple Notification Service (SNS)

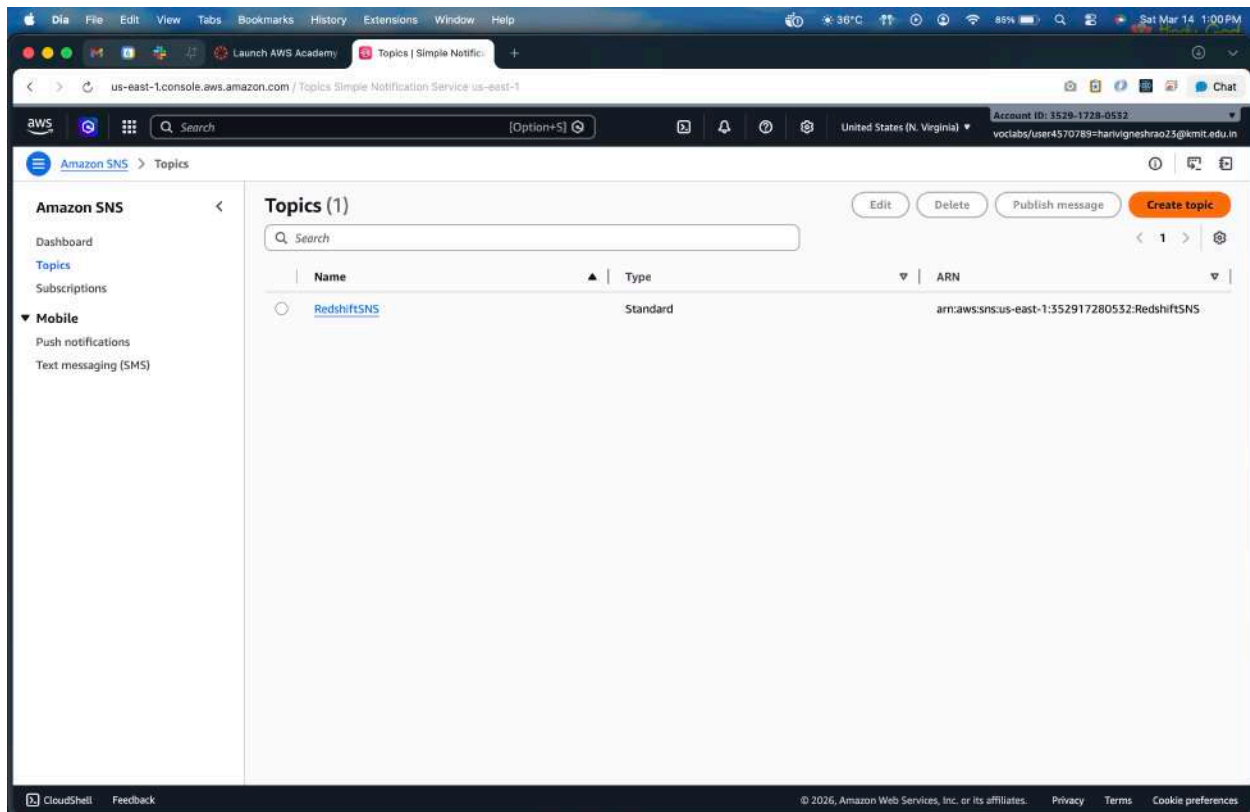
A) Create SNS Topic and Send Message to Email

We will:

1. Create an SNS Topic
2. Create an Email Subscription
3. Confirm the subscription
4. Publish a test message

Step 1: Create SNS Topic

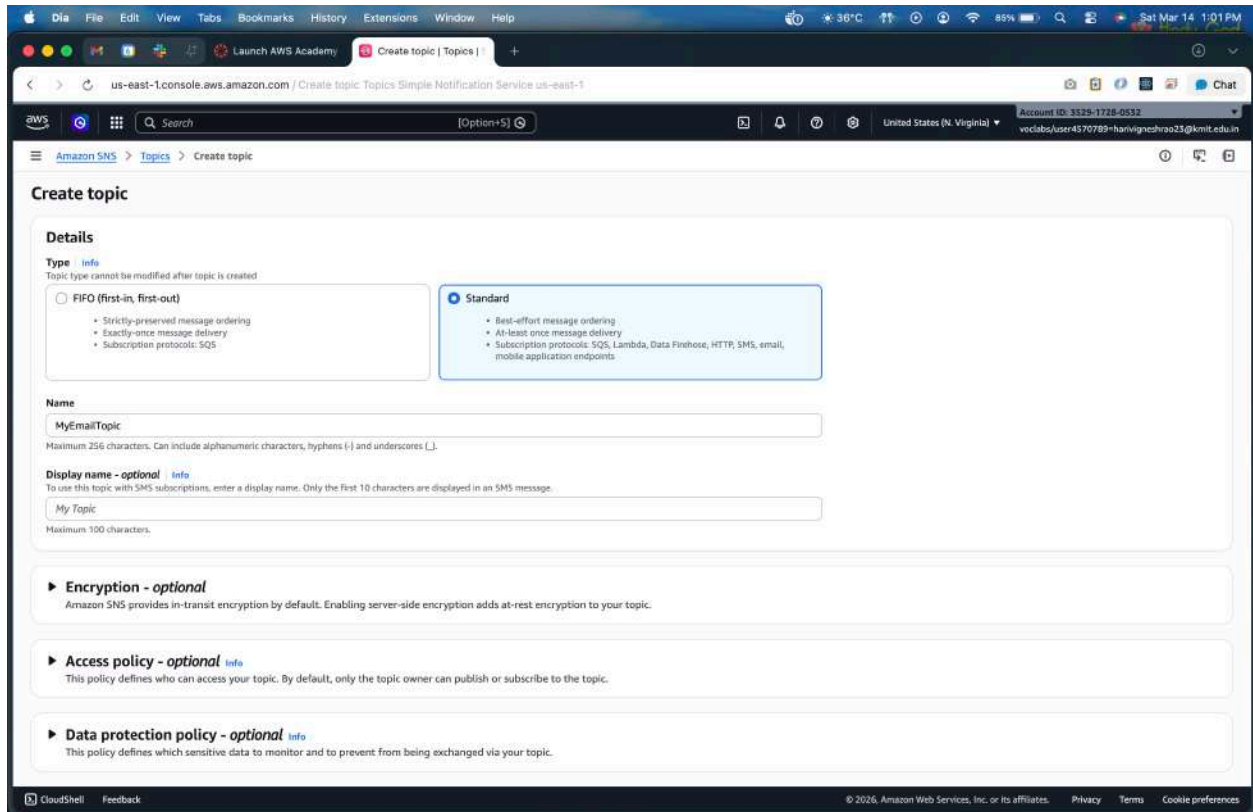
1. Login to **AWS Console**
2. In search bar → Type **SNS**
3. Open **Amazon SNS (Simple Notification Service)**
4. In left panel → Click **Topics**
5. Click **Create topic**



Configure:

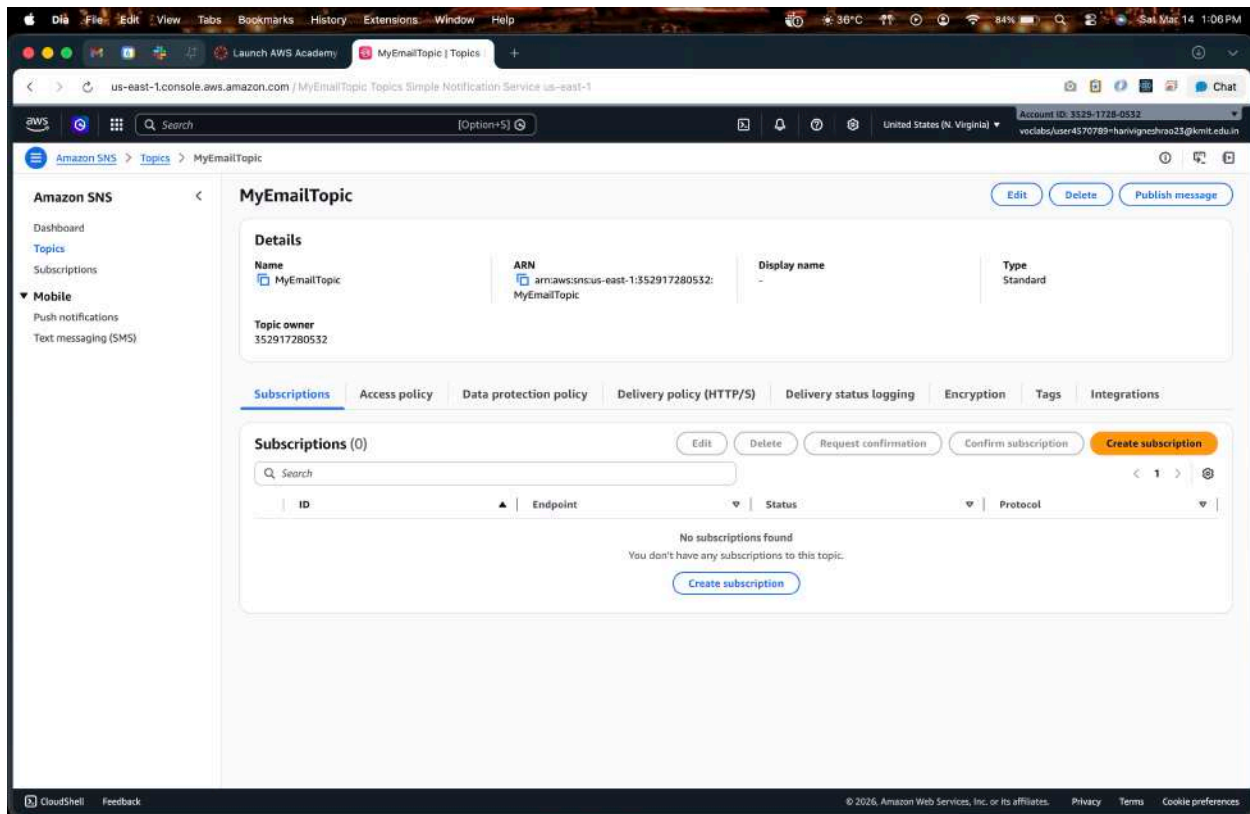
- **Type** → Select **Standard**
- **Name** → MyEmailTopic
- Leave other settings default (unless required)

6. Click **Create topic**



Step 2: Create Email Subscription

1. Open your created topic
2. Go to **Subscriptions** tab
3. Click **Create subscription**



Configure:

- **Protocol** → Select **Email**
- **Endpoint** → Enter your email address

4. Click **Create subscription**

The screenshot shows the AWS Management Console interface for creating a new subscription. The browser address bar indicates the URL is `us-east-1.console.aws.amazon.com / Create subscription Subscriptions Simple Notification Service us-east-1`. The page title is "Create subscription".

Details

Topic ARN
A text input field containing `arn:aws:sns:us-east-1:352917280532:MyEmailTopic`.

Protocol
A dropdown menu with "Email" selected. The description below reads: "The type of endpoint to subscribe".

Endpoint
A text input field containing `harivigneshrao151@gmail.com`. The description below reads: "An email address that can receive notifications from Amazon SNS."

A light blue informational box states: "After your subscription is created, you must confirm it. [Info](#)."

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

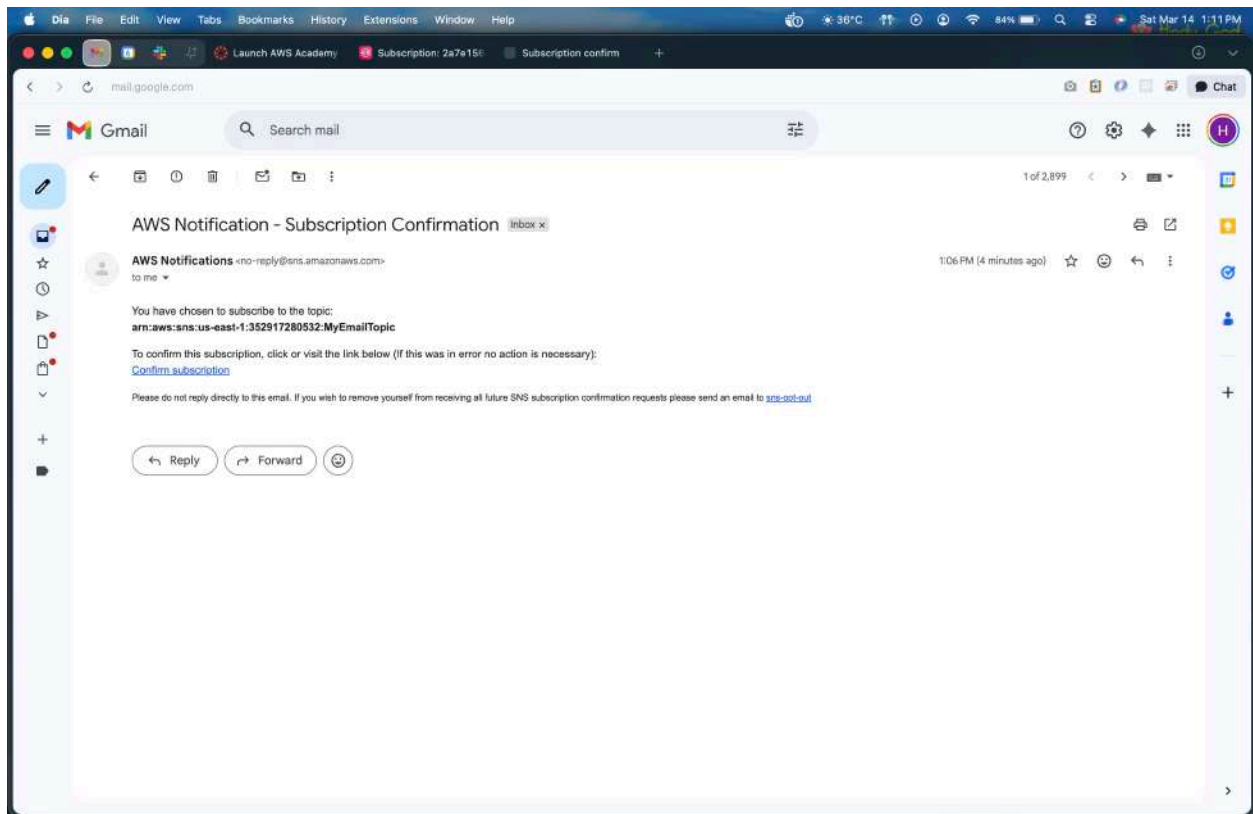
Redrive policy (dead-letter queue) - optional [Info](#)
Send undeliverable messages to a dead-letter queue.

At the bottom right, there are two buttons: "Cancel" and "Create subscription".

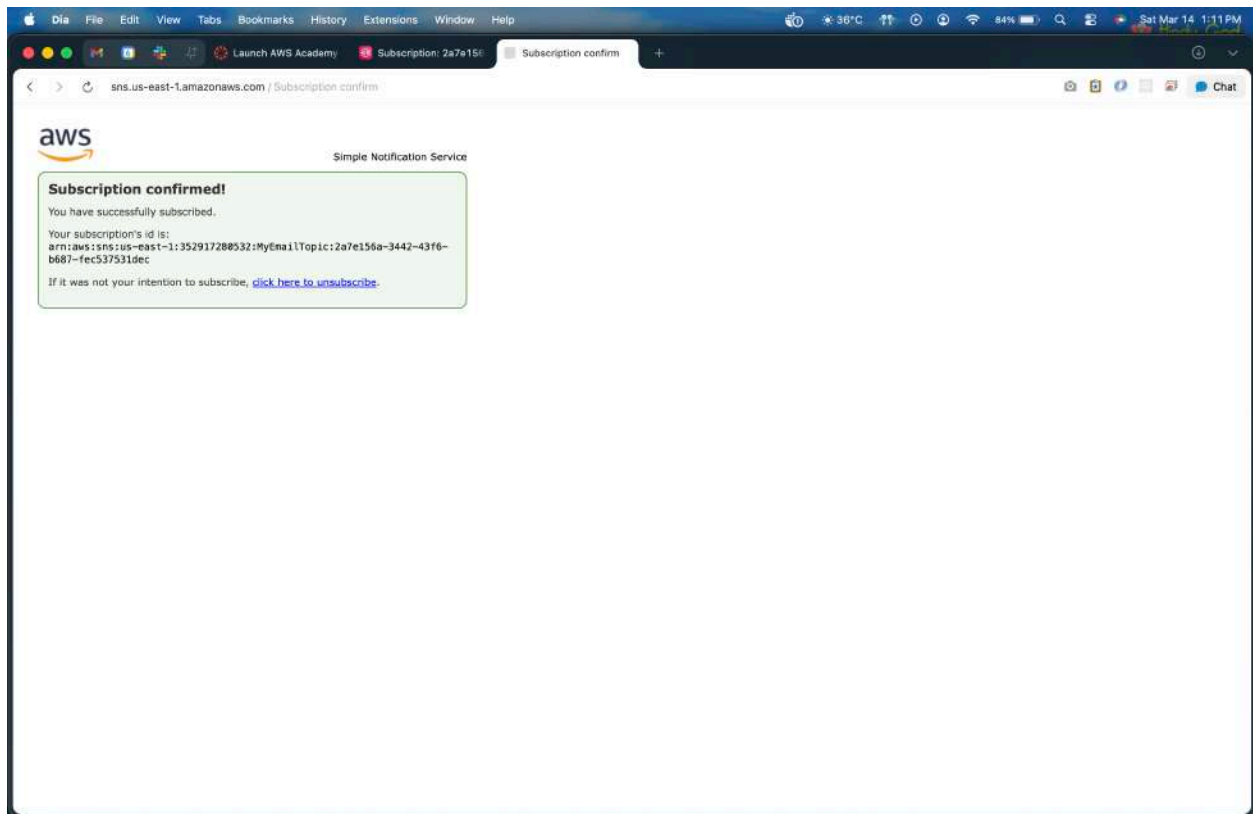
The footer of the console shows "CloudShell", "Feedback", and copyright information: "© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences".

Step 3: Confirm Subscription

- Go to your email inbox
- You will receive a mail from **Amazon SNS**
- Click **Confirm subscription**

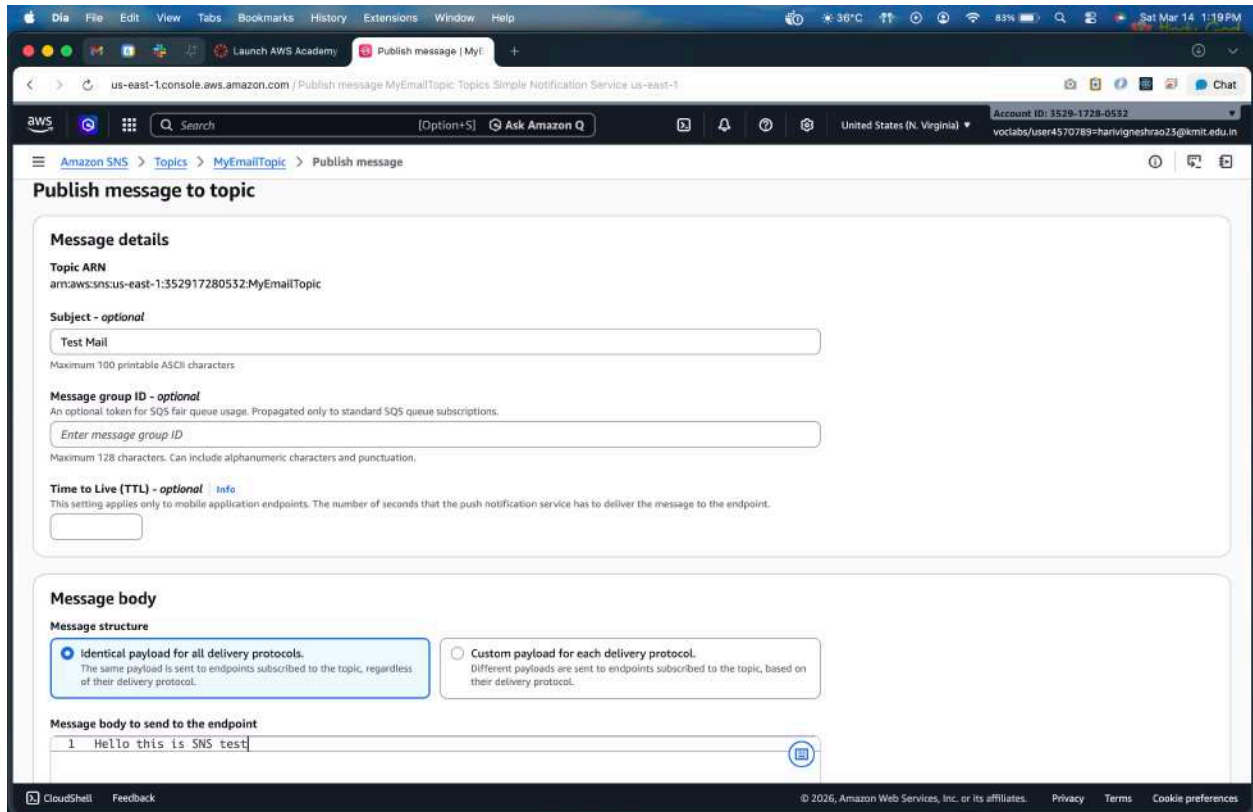


Now your email is successfully subscribed 🎉

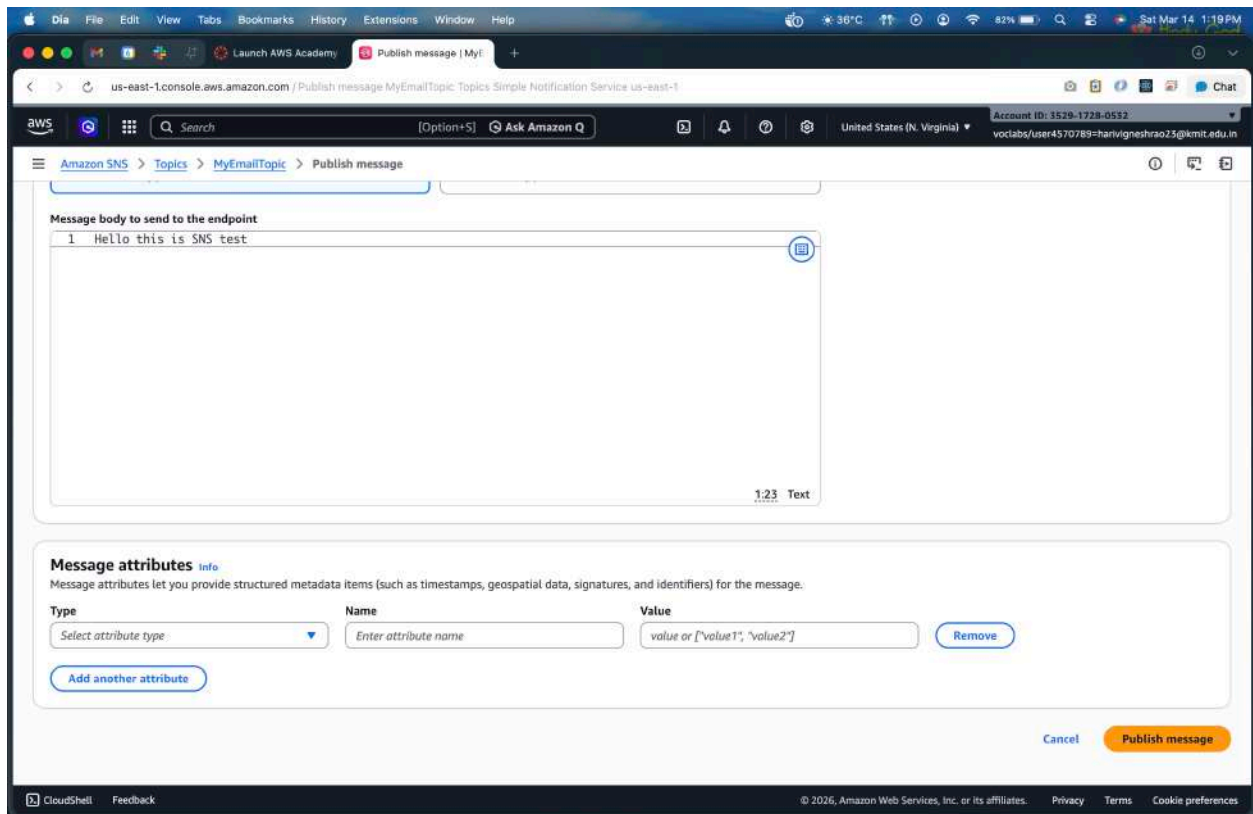


Step 4: Publish Test Message

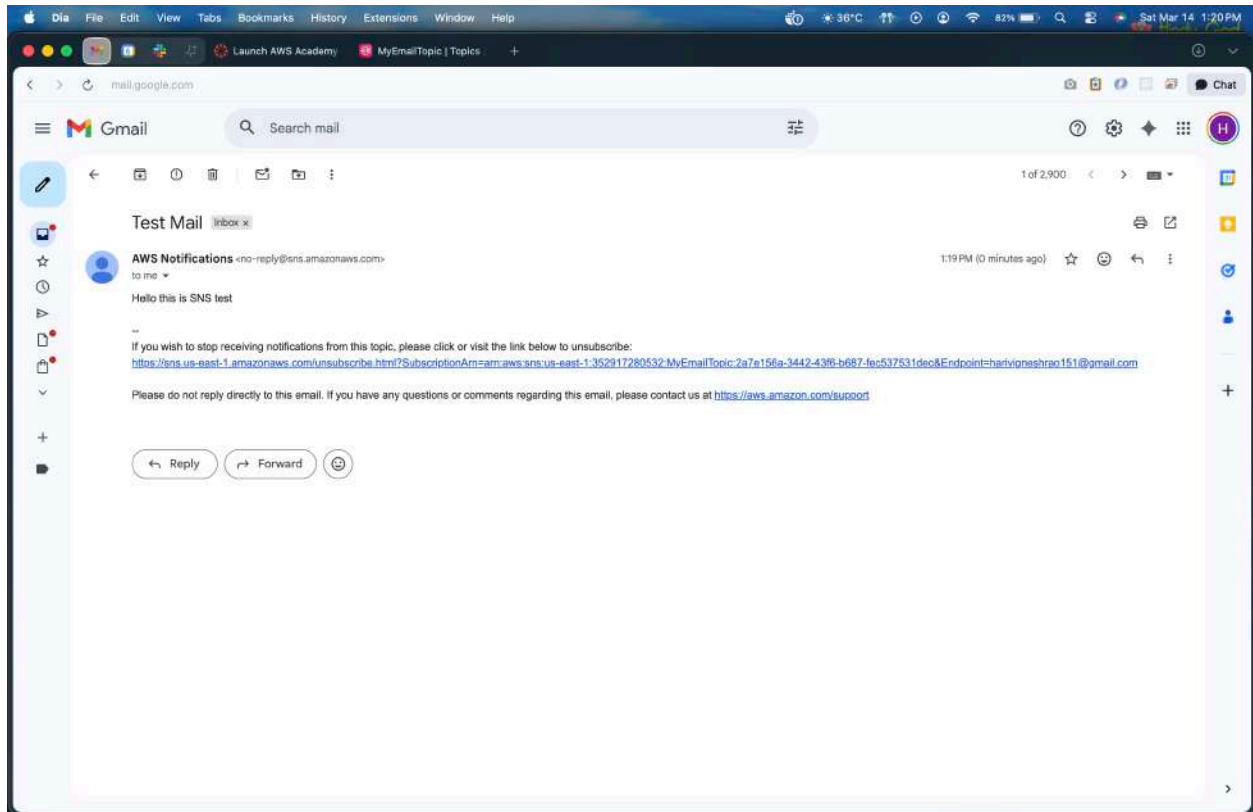
1. Open your SNS topic
2. Click **Publish message**
3. Add:
 - **Subject** → Test Mail
 - **Message body** → Hello this is SNS test



4. Click **Publish message**



You will receive email notification.



B) Trigger Email When S3 Object is Uploaded (S3 → SNS → Email)

Now we extend this setup:

Flow:

S3 Bucket → Event Notification → SNS Topic → Email

Step 1: Create S3 Bucket

1. Search → **S3**
2. Click **Create bucket**

Configure:

- **Bucket name** → my-upload-bucket-2026
 - Choose Region (Important: Same region as SNS topic)
 - Keep default settings (unless specific requirement)
3. Click **Create bucket**

us-east-1.console.aws.amazon.com / Create S3 bucket S3 us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

Amazon S3 > Buckets > Create bucket

General configuration

AWS Region

US East (N. Virginia) us-east-1

Bucket type

General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace

Choose the namespace where you want to create your bucket. [Learn more](#)

Global namespace

By default, S3 creates general purpose buckets in the global namespace.

Account Regional namespace (recommended)

General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name

my-upload-bucket-2027-28

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

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us-east-1.console.aws.amazon.com / Create S3 bucket S3 us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

Amazon S3 > Buckets > Create bucket

No tags associated with this bucket.

Add new tag

You can add up to 50 tags.

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Server-side encryption with AWS Key Management Service keys (SSE-KMS)

Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

Disable

Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket

CloudShell

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Step 2: Create SNS Topic (If not already created)

You can reuse the topic from Part A
OR create new topic like:

- Topic name → MyEmailTopic

Make sure email subscription is confirmed.

The screenshot shows the AWS Management Console interface for creating a new SNS topic. The browser address bar indicates the URL is `us-east-1.console.aws.amazon.com / Create topic Topics Simple Notification Service us-east-1`. The page title is "Create topic".

Details

Type [Info](#)
Topic type cannot be modified after topic is created.

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least-once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

MyEmailTopic

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

My Topic

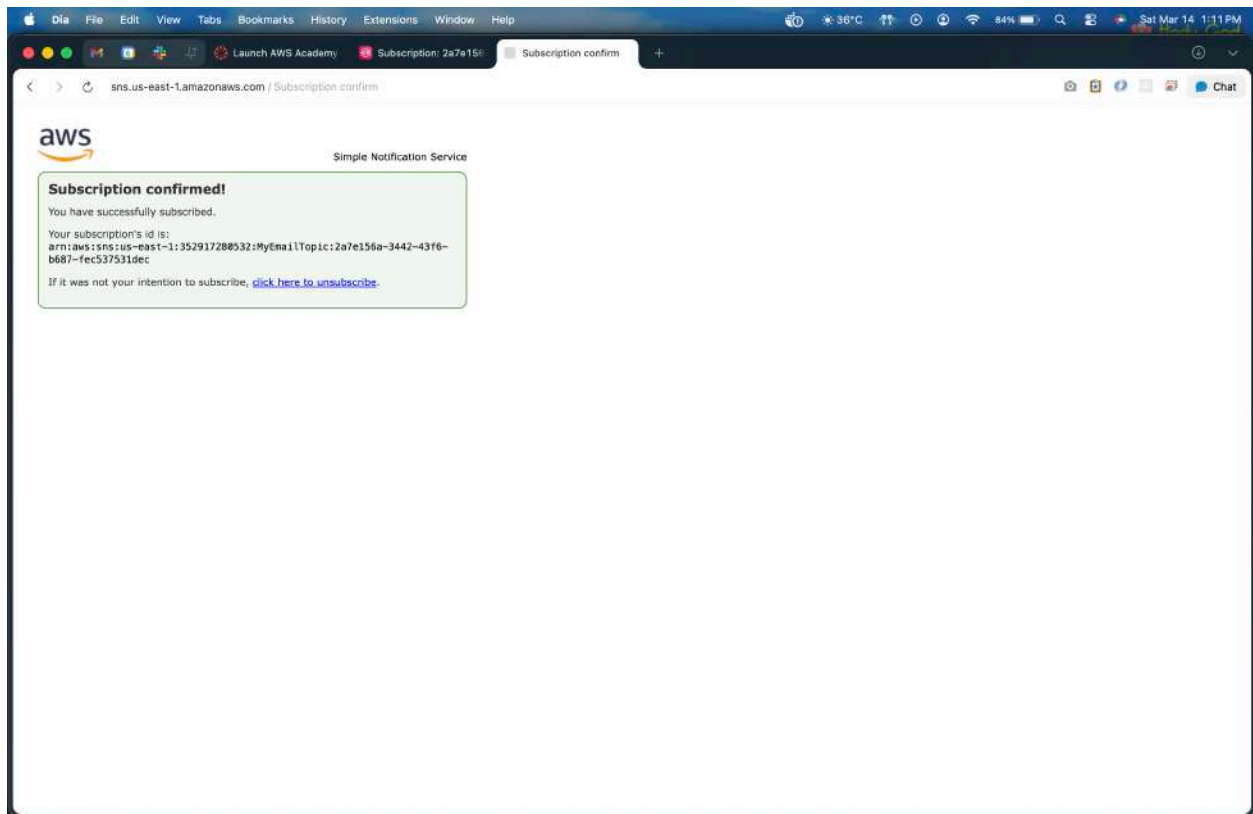
Maximum 100 characters.

Encryption - optional
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

Access policy - optional [Info](#)
This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

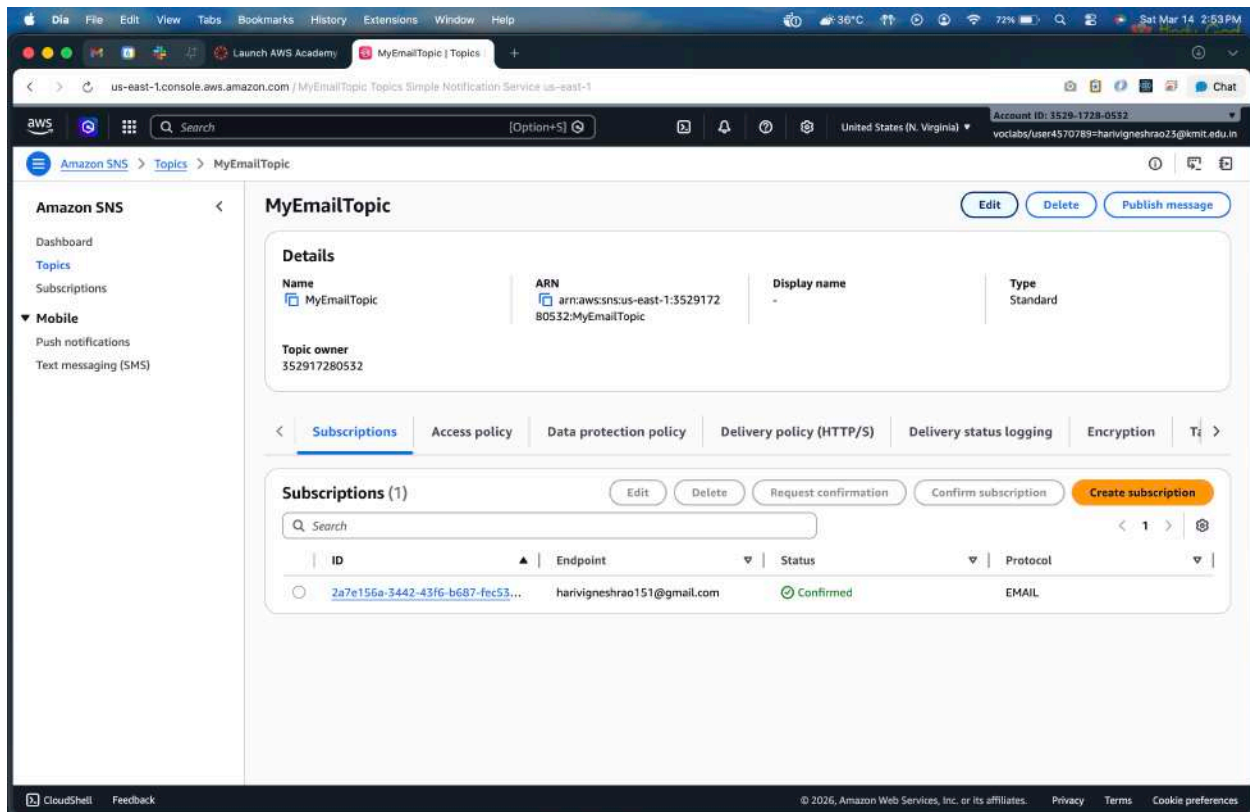
Data protection policy - optional [Info](#)
This policy defines which sensitive data to monitor and to prevent from being exchanged via your topic.

At the bottom of the console, there is a footer with "© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences".



Step 3: Allow S3 to Publish to SNS (Access Policy)

1. Open SNS Topic
2. Go to **Access policy**
3. Click **Edit**



If using Basic Editor:

Access Policy that needs to be configured on SNS:

a) replace SNS topic ARN

b) Replace s3 bucket ARN

c) Replace account

{

"Version": "2008-10-17",

"Id": "__default_policy_ID",

"Statement": [

{

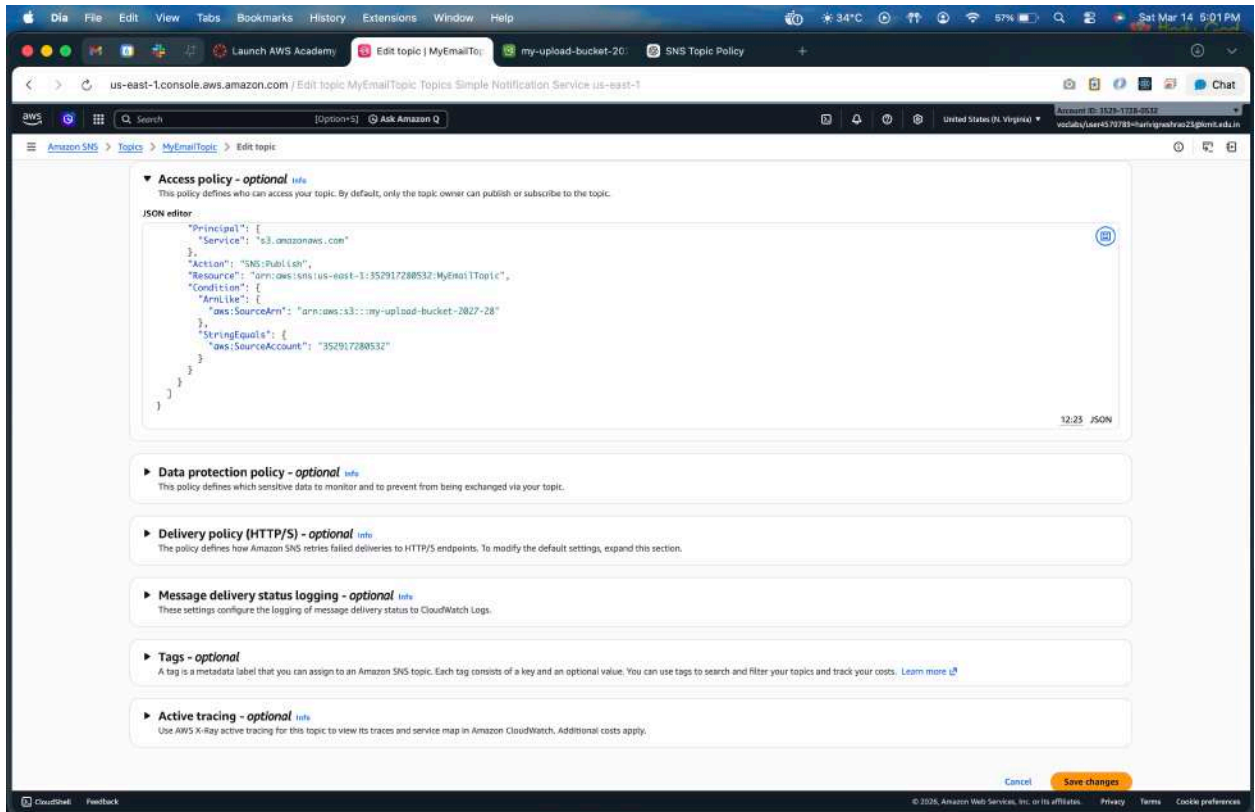
"Sid": "__default_statement_ID",

"Effect": "Allow",

```
"Principal": {  
  "AWS": "*"   
},  
  
"Action": [  
  "SNS:Publish",  
  
  "SNS:RemovePermission",  
  
  "SNS:SetTopicAttributes",  
  
  "SNS:DeleteTopic",  
  
  "SNS:ListSubscriptionsByTopic",  
  
  "SNS:GetTopicAttributes",  
  
  "SNS:AddPermission",  
  
  "SNS:Subscribe"  
],  
  
"Resource": "arn:aws:sns:us-east-1:352917280532:MyEmailTopic",  
  
"Condition": {  
  "StringEquals": {  
    "AWS:SourceAccount": "352917280532"  
  }  
}  
  
},  
  
{  
  "Sid": "AllowS3Publish",  
  
  "Effect": "Allow",
```

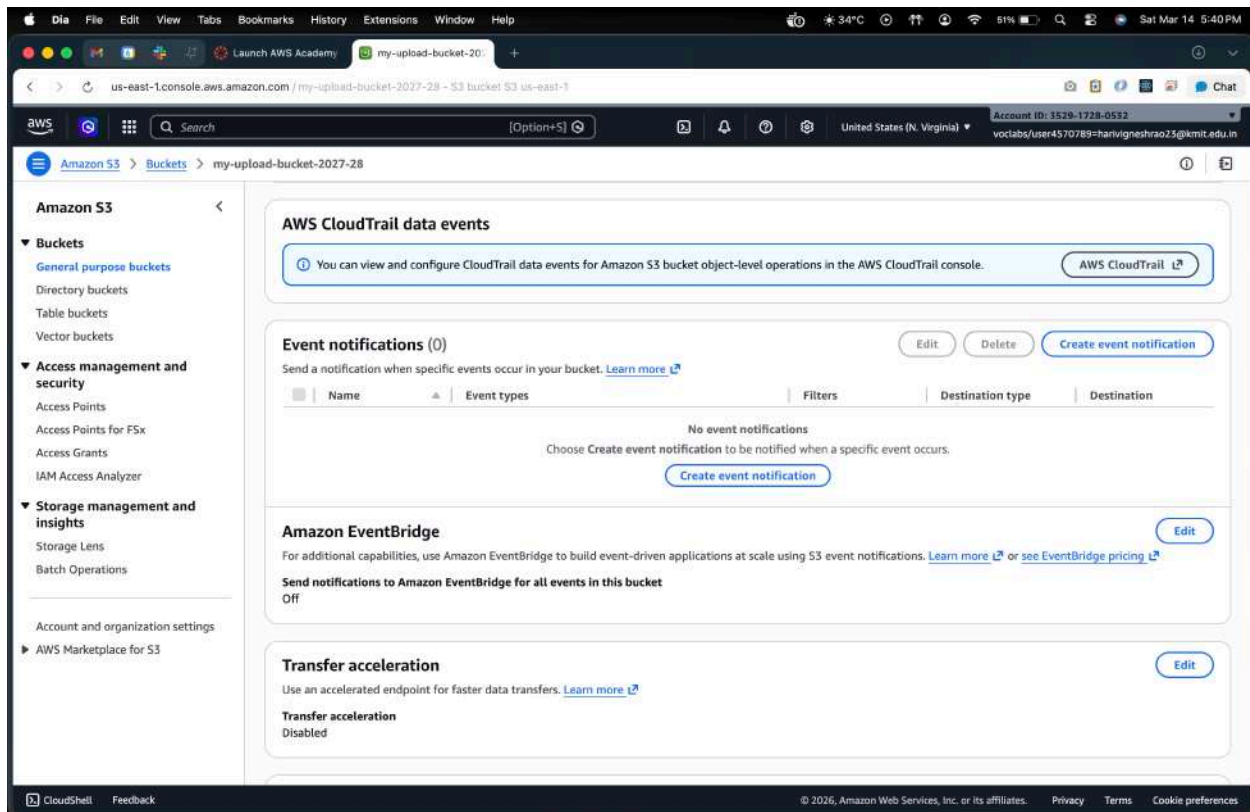
```
"Principal": {  
  "Service": "s3.amazonaws.com"  
},  
"Action": "SNS:Publish",  
"Resource": "arn:aws:sns:us-east-1:352917280532:MyEmailTopic",  
"Condition": {  
  "ArnLike": {  
    "aws:SourceArn": "arn:aws:s3:::my-upload-bucket-2027-28"  
  },  
  "StringEquals": {  
    "aws:SourceAccount": "352917280532"  
  }  
}  
}  
]  
}
```

(Modern AWS console often auto-generates this when configuring from S3)



Step 4: Configure S3 Event Notification

1. Open your S3 bucket
2. Go to **Properties**
3. Scroll to **Event notifications**
4. Click **Create event notification**



Configure:

- **Event name** → UploadNotification
- **Event types** → Select:
 - All object create events
- **Destination:**
 - Select **SNS topic**
 - Choose your topic (S3UploadNotification)

5. Click **Save changes**

us-east-1.console.aws.amazon.com / Create event notification - S3 bucket my-upload-bucket-2027-28 S3 us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao.23@kmit.edu.in

Amazon S3

Buckets

my-upload-bucket-2027-28

Create event notification

Create event notification [info](#)

To enable notifications, you must first add a notification configuration that identifies the events you want Amazon S3 to publish and the destinations where you want Amazon S3 to send the notifications.

General configuration

Event name

UploadNotification

Event name can contain up to 255 characters.

Prefix - optional

Limit the notifications to objects with key starting with specified characters.

images/

Suffix - optional

Limit the notifications to objects with key ending with specified characters.

.jpg

Event types

Specify at least one event for which you want to receive notifications. For each group, you can choose an event type for all events, or you can choose one or more individual events.

Object creation

☒ All object create events
s3:ObjectCreated:*

☐ Put
s3:ObjectCreated:Put

☐ Post
s3:ObjectCreated:Post

☐ Copy
s3:ObjectCreated:Copy

CloudShell

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us-east-1.console.aws.amazon.com / Create event notification - S3 bucket my-upload-bucket-2027-28 S3 us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao.23@kmit.edu.in

Amazon S3

Buckets

my-upload-bucket-2027-28

Create event notification

☐ Intelligent-Tiering archive events
s3:IntelligentTiering

Destination

Before Amazon S3 can publish messages to a destination, you must grant the Amazon S3 principal the necessary permissions to call the relevant API to publish messages to an SNS topic, an SQS queue, or a Lambda function. [Learn more](#)

Destination

Choose a destination to publish the event. [Learn more](#)

☐ Lambda function
Run a Lambda function script based on S3 events.

☒ SNS topic
Fanout messages to systems for parallel processing or directly to people.

☐ SQS queue
Send notifications to an SQS queue to be read by a server.

Specify SNS topic

☒ Choose from your SNS topics

☐ Enter SNS topic ARN

SNS topic

MyEmailTopic

Cancel

Save changes

CloudShell

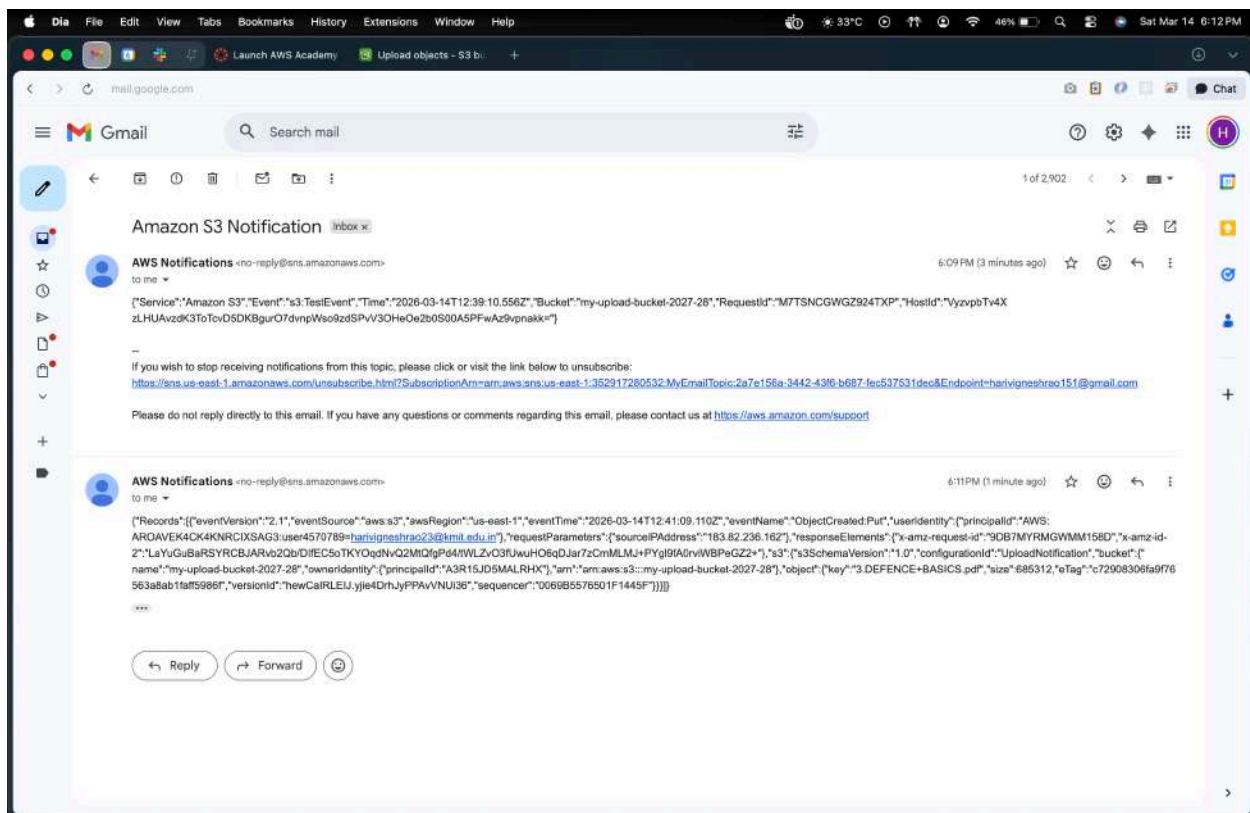
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Step 5: Test It

1. Upload any file into the S3 bucket
2. After upload completes:
 - S3 triggers event
 - SNS receives publish request
 - SNS sends email

You will receive mail about object upload.



Simple Queue Service (SQS)

2) Create SQS Queue and Send Message

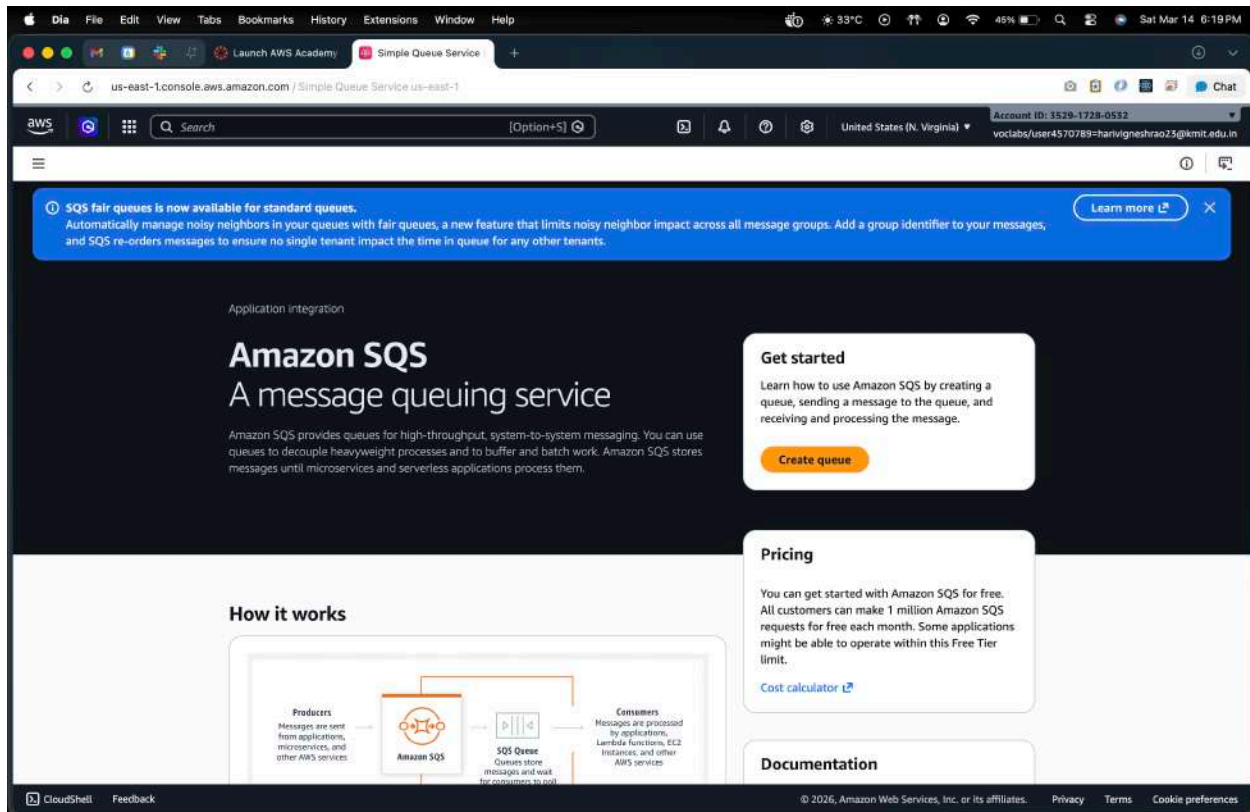
We will:

1. Create an SQS Queue
2. Send a message to the Queue
3. Receive the message from the Queue

Step 1: Create SQS Queue

1. Login to **AWS Console**
2. In the **search bar** → **Type SQS**

3. Open **Amazon Simple Queue Service**
4. In the dashboard → Click **Create queue**



Configure:

- **Type** → **Select Standard**
 - **Name** → **MyQueue**
 - Leave other settings as **default**
5. Click **Create queue**

Your **SQS queue** is now created successfully.

us-east-1.console.aws.amazon.com / Create queue Simple Queue Service us-east-1

Amazon SQS > Queues > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

☒ Standard [Info](#)

At-least-once delivery, message ordering isn't preserved

- At-least once delivery
- Best-effort ordering

☐ FIFO [Info](#)

First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

☐ You can't change the queue type after you create a queue.

Name

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration [Info](#)

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)

 Seconds

Should be between 0 seconds and 12 hours.

Message retention period [Info](#)

 Days

Should be between 1 minute and 14 days.

Delivery delay [Info](#)

 Seconds

Should be between 0 seconds and 15 minutes.

Maximum message size [Info](#)

 KiB

Should be between 1 KiB and 1024 KiB.

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us-east-1.console.aws.amazon.com / Create queue Simple Queue Service us-east-1

Amazon SQS > Queues > Create queue

Redrive allow policy - Optional [Info](#)

Identify which source queues can use this queue as the dead-letter queue.

Select which source queues can use this queue as the dead-letter queue.

☒ Disabled

☐ Enabled

Dead-letter queue - Optional [Info](#)

Send undeliverable messages to a dead-letter queue.

Set this queue to receive undeliverable messages.

☒ Disabled

☐ Enabled

Tags - Optional [Info](#)

A tag is a label assigned to an AWS resource. Use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Remove

Add new tag

You can add 49 more tags.

Cancel

Create queue

CloudShell

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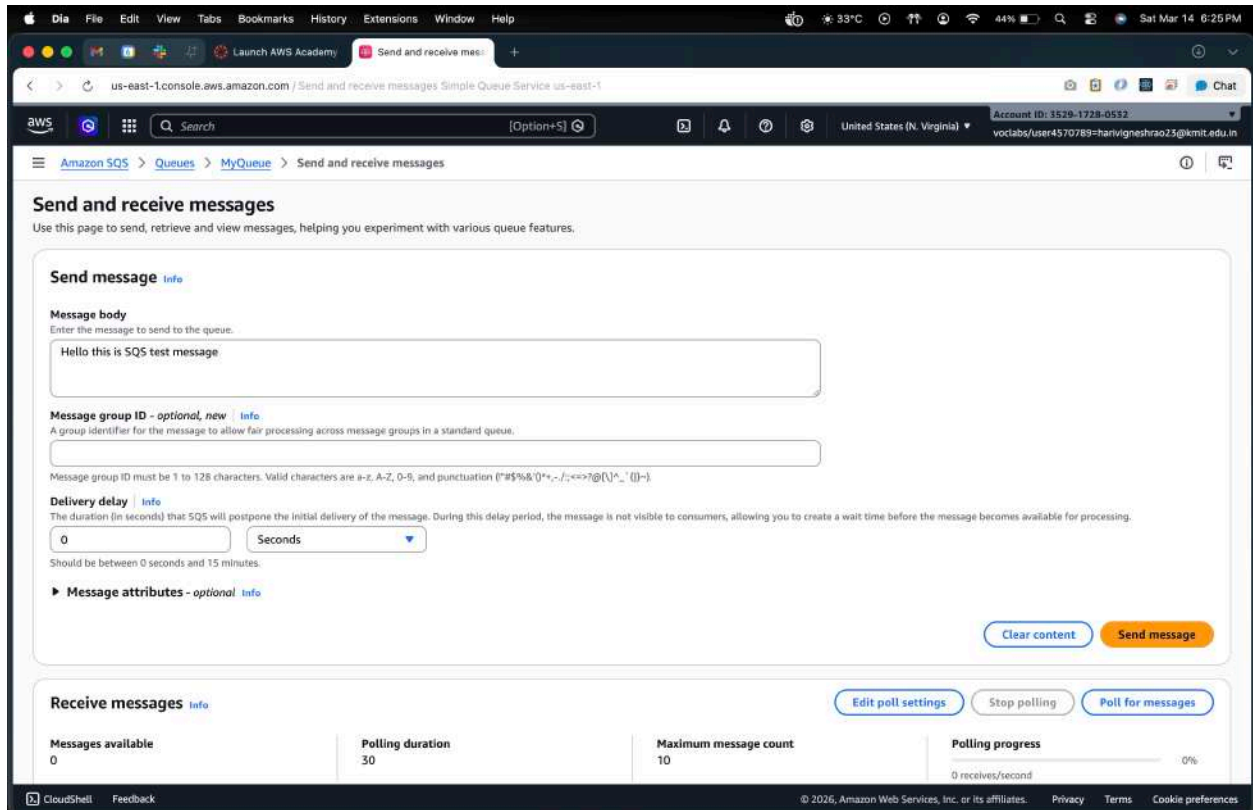
Step 2: Send Message to Queue

1. Open the **MyQueue** queue
2. Click **Send and receive messages**
3. In the **Message body**, type:

Hello this is SQS test message

4. Click **Send message**

Your message is now stored in the queue.



Step 3: Receive Message from Queue

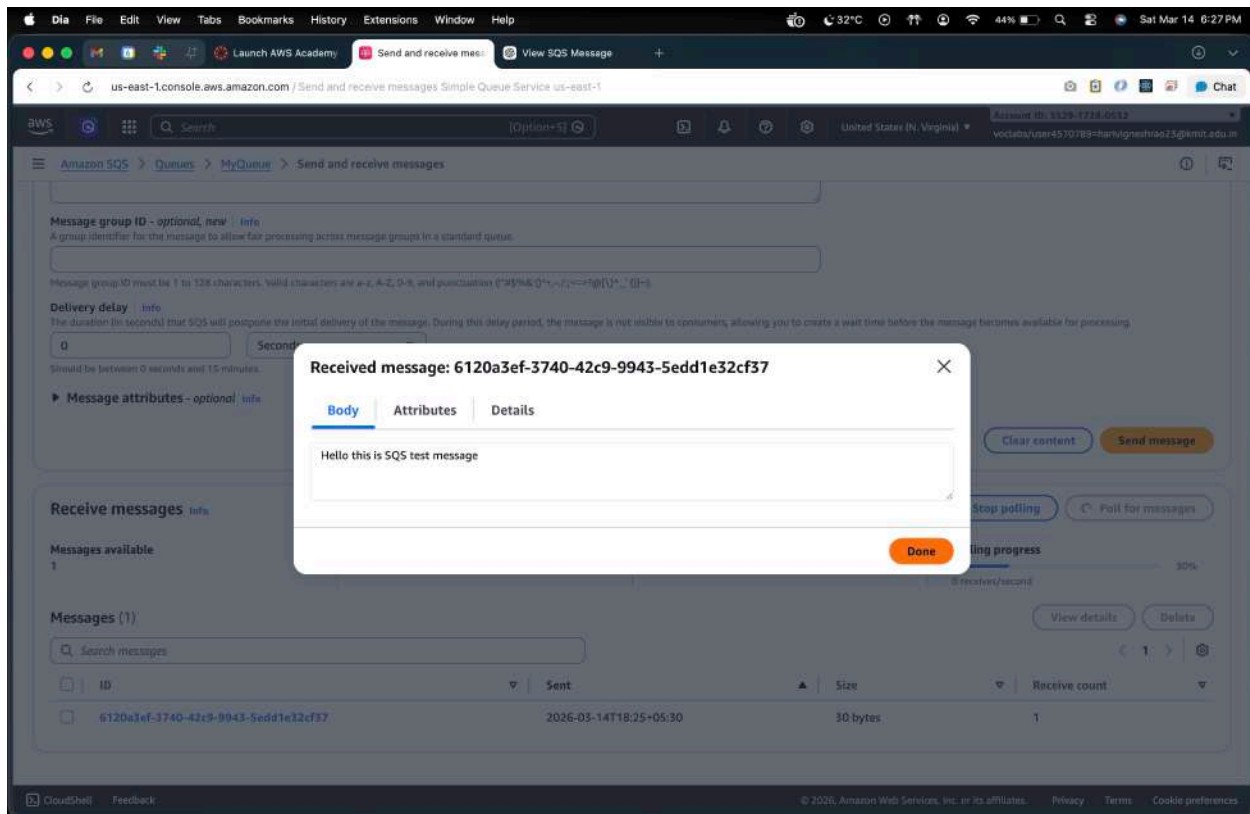
1. In the same page (**Send and receive messages**)
2. Click **Poll for messages**

You will see the message displayed.

Example output:

Hello this is SQS test message

3. After processing, click **Delete message** (optional)



Result:

- SQS queue created
- Message sent successfully
- Message received from queue

3) S3 Bucket Event → SNS → SQS → Lambda Processing

S3 bucket event (like file upload) triggers SNS, which sends the notification to SQS. Then **Lambda acts as the consumer and processes the message.**

Services used:

- Amazon S3
- Amazon Simple Notification Service
- Amazon Simple Queue Service
- AWS Lambda

S3 → SNS → SQS → Lambda Architecture

S3 Bucket (File Upload)



SNS Topic



SQS Queue

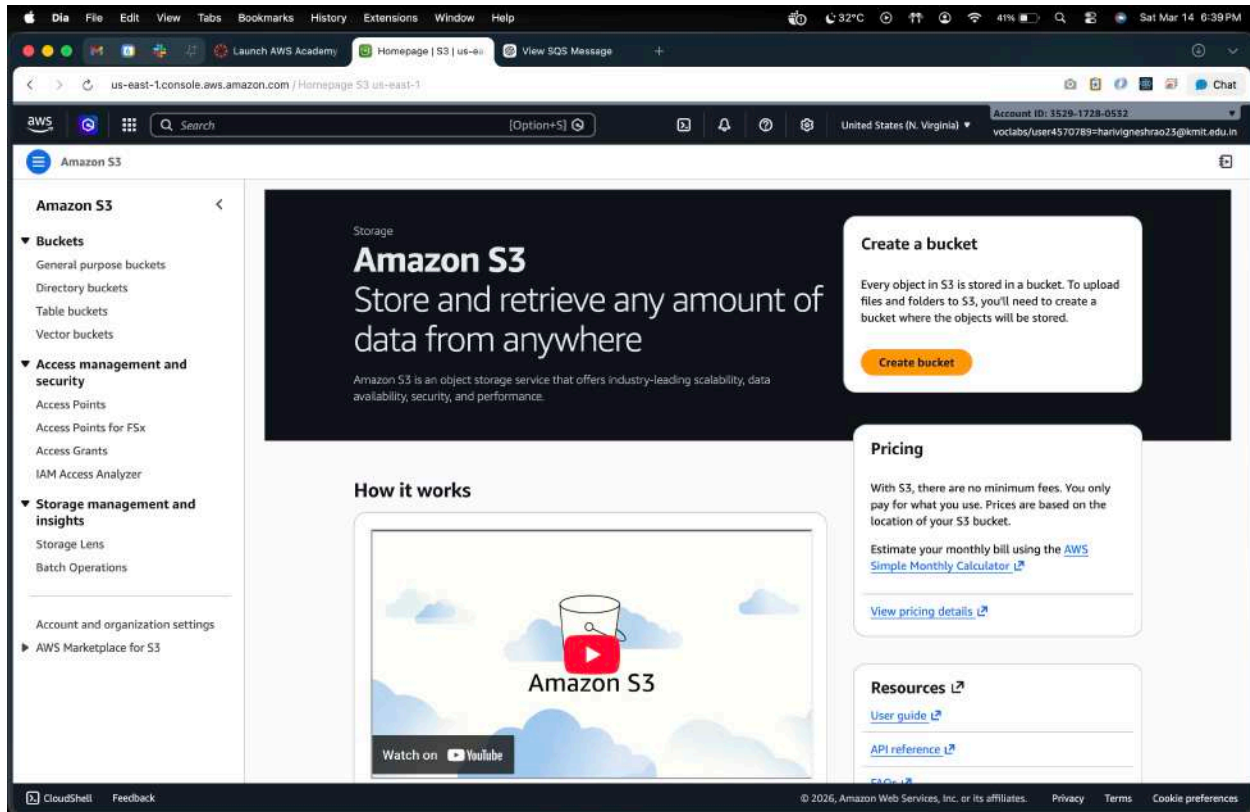


Lambda Function (Consumer)

When a file is uploaded, S3 sends an event notification to SNS, which forwards the message to SQS. The Lambda function then reads and processes the message from SQS.

Step 1: Create S3 Bucket

1. Login to AWS Console
2. Search **S3**
3. Open **Amazon S3**
4. Click **Create bucket**

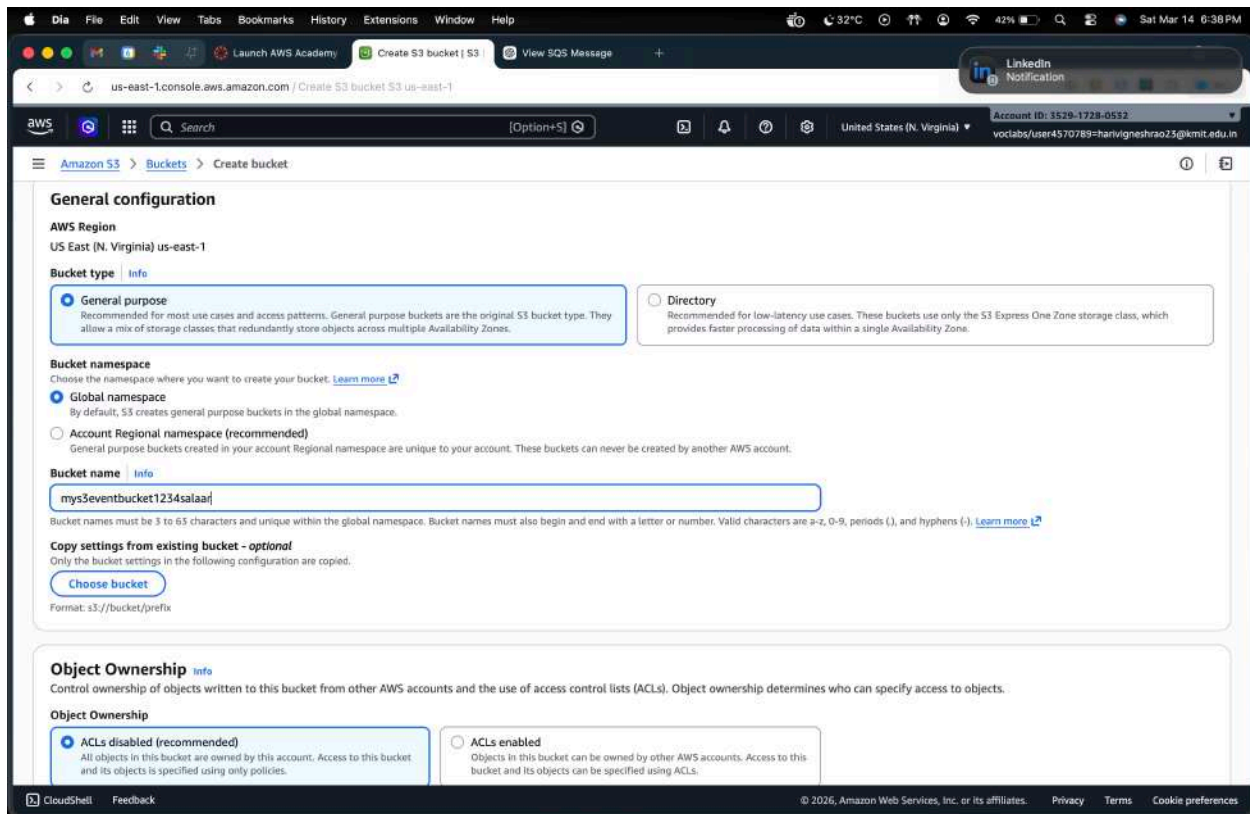


Configure:

- Bucket name → **mys3eventbucket123** (must be unique)
- Region → choose your region

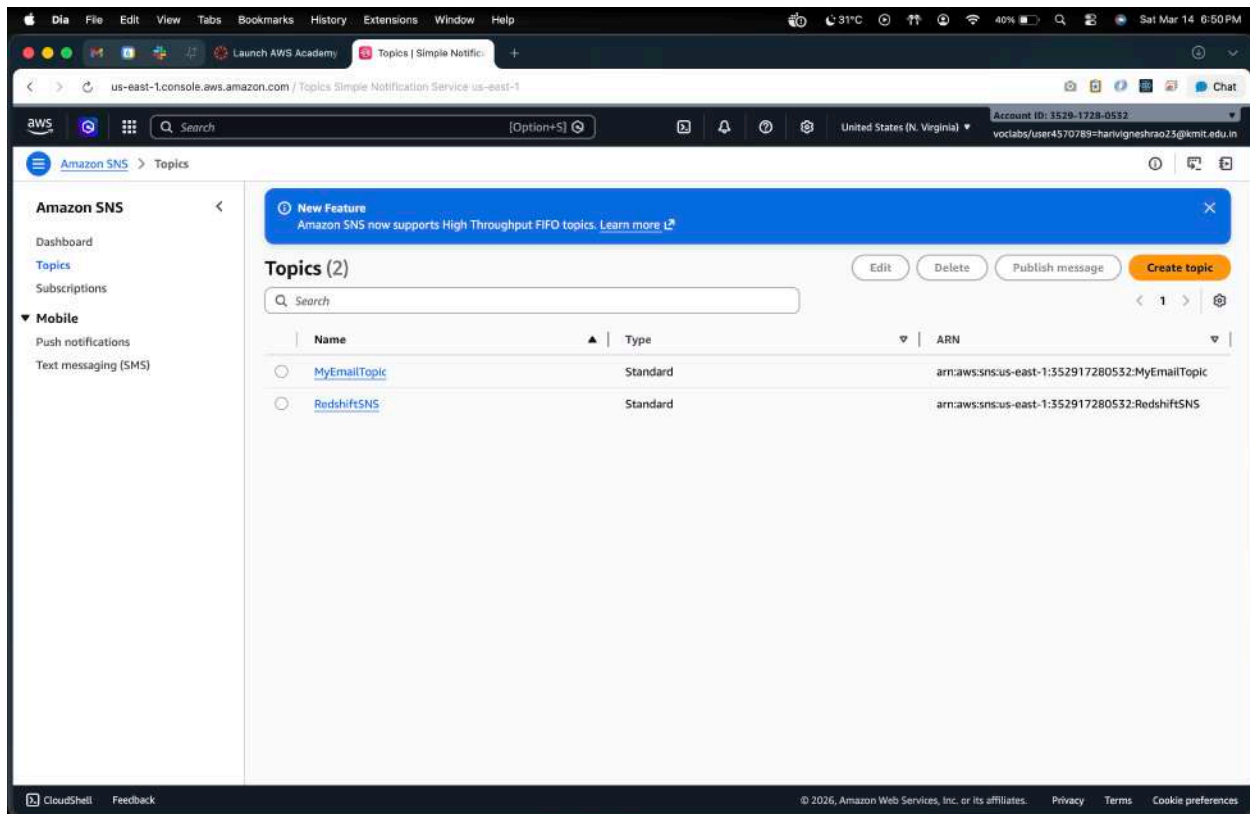
Leave other settings default.

5. Click **Create bucket**



Step 2: Create SNS Topic

1. Search SNS in AWS Console
2. Open **Amazon SNS**
3. Click **Topics** → **Create topic**



Configure:

- Type → **Standard**
- Name → **MyS3SNSTopic**

4. Click **Create topic**

Copy the **SNS Topic ARN**.

us-east-1.console.aws.amazon.com / Create topic Topics Simple Notification Service us-east-1

Amazon SNS > Topics > Create topic

Create topic

Details

Type [info](#)

Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

MySSNSTopic

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [info](#)

To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

My Topic

Maximum 100 characters.

► Encryption - optional

Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► Access policy - optional [info](#)

This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

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us-east-1.console.aws.amazon.com / Create topic Topics Simple Notification Service us-east-1

Amazon SNS > Topics > Create topic

New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

► Access policy - optional [info](#)

This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

► Archive policy - new, optional [info](#)

This policy tells Amazon SNS how long to store your messages so that they can be resent to a subscription. By default, Amazon SNS does not retain your messages.

► Message delivery status logging - optional [info](#)

These settings configure the logging of message delivery status to CloudWatch Logs.

► Tags - optional

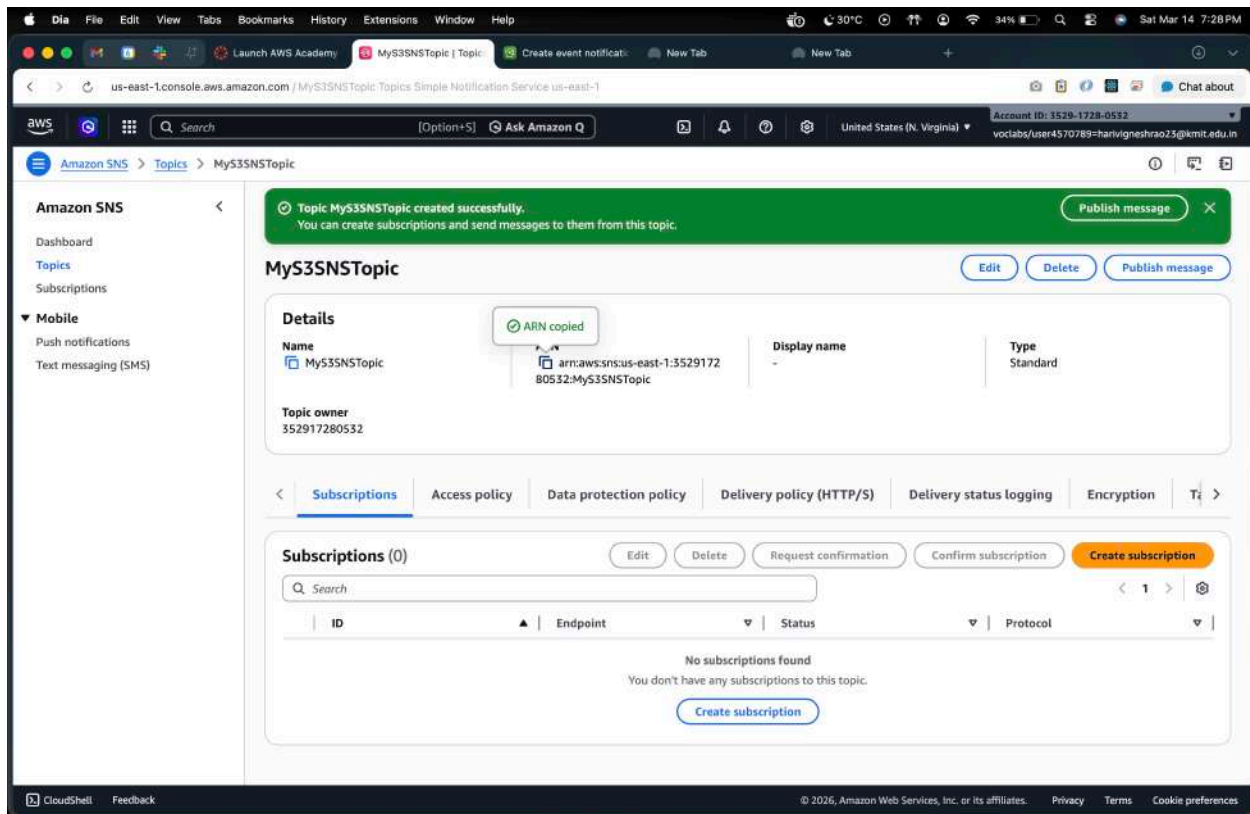
A tag is a metadata label that you can assign to an Amazon SNS topic. Each tag consists of a key and an optional value. You can use tags to search and filter your topics and track your costs. [Learn more](#)

► Active tracing - optional [info](#)

Use AWS X-Ray active tracing for this topic to view its traces and service map in Amazon CloudWatch. Additional costs apply.

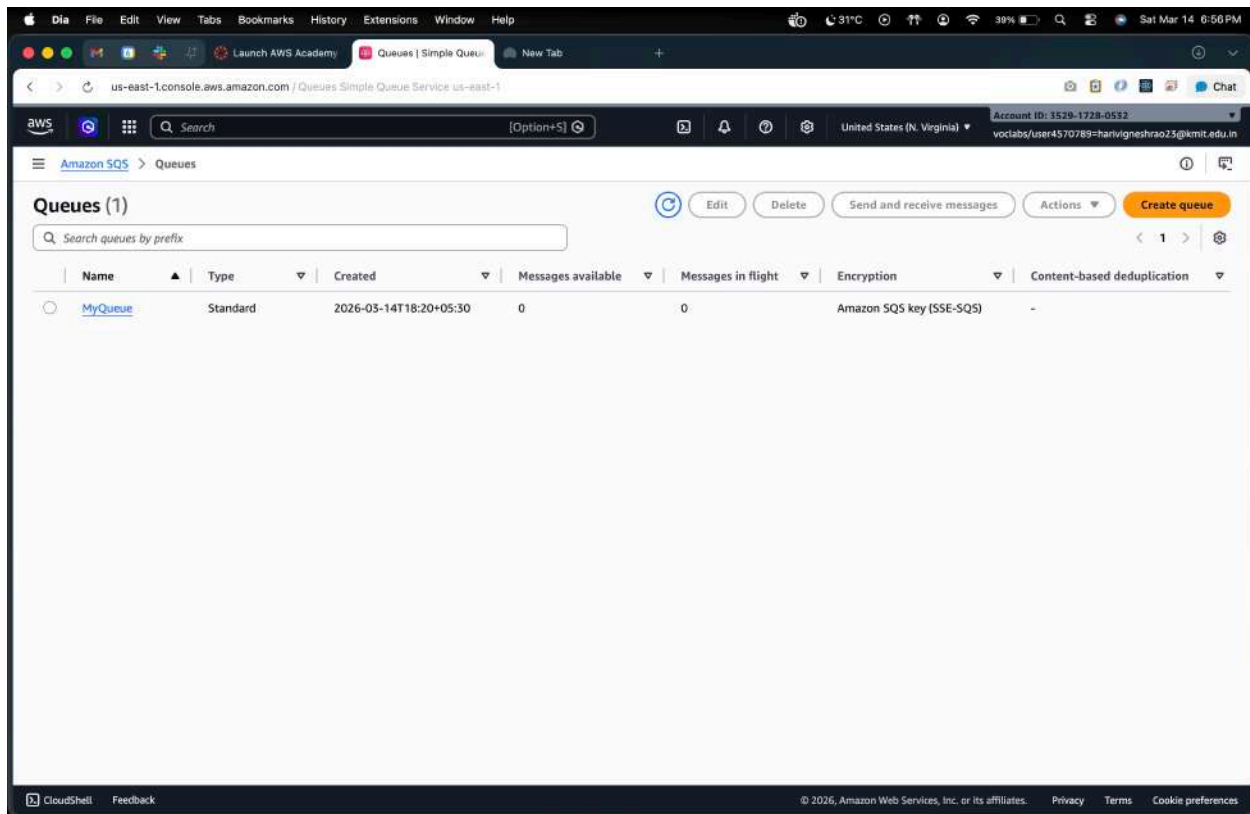
Cancel Create topic

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Step 3: Create SQS Queue

1. Search **SQS** in AWS Console
2. Open **Amazon SQS**
3. Click **Create queue**



Configure:

- Type → **Standard**
- Name → **MyS3Queue**

4. Click **Create queue**

Copy the **Queue ARN**.

us-east-1.console.aws.amazon.com / Create queue | Simple Queue Service us-east-1

Amazon SQS > Queues > Create queue

Create queue

Details

Type
Choose the queue type for your application or cloud infrastructure.

☒ **Standard** [Info](#)
At-least-once delivery, message ordering isn't preserved

- At-least once delivery
- Best-effort ordering

☐ **FIFO** [Info](#)
First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

Name

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)

Seconds

Should be between 0 seconds and 12 hours.

Message retention period [Info](#)

Days

Should be between 1 minute and 14 days.

Delivery delay [Info](#)

Seconds

Should be between 0 seconds and 15 minutes.

Maximum message size [Info](#)

KiB

Should be between 1 KiB and 1024 KiB.

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us-east-1.console.aws.amazon.com / Create queue | Simple Queue Service us-east-1

Amazon SQS > Queues > Create queue

Redrive allow policy - Optional

Identify which source queues can use this queue as the dead-letter queue.

Select which source queues can use this queue as the dead-letter queue.

☒ Disabled

☐ Enabled

Dead-letter queue - Optional

Send undeliverable messages to a dead-letter queue.

Set this queue to receive undeliverable messages.

☒ Disabled

☐ Enabled

Tags - Optional

A tag is a label assigned to an AWS resource. Use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Enter key	Enter value	<button>Remove</button>

Add new tag

You can add 49 more tags.

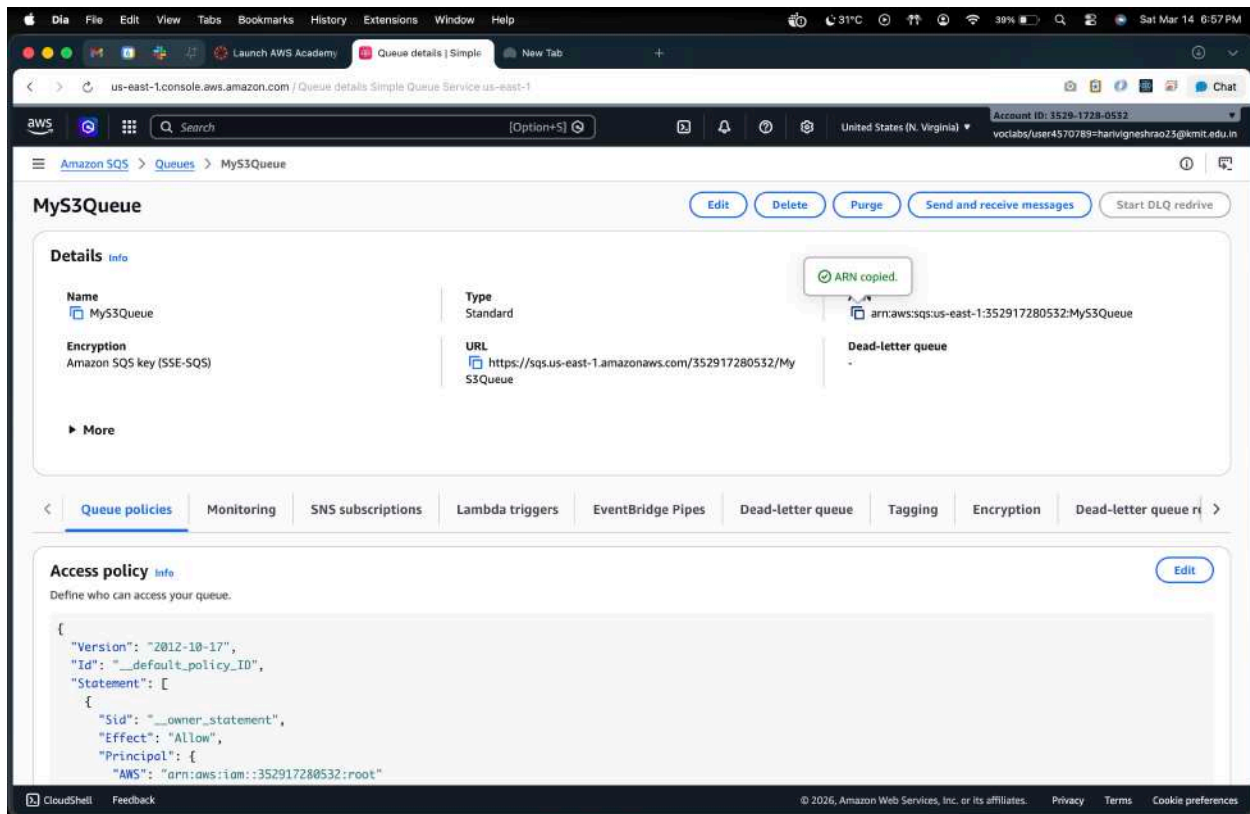
Cancel

Create queue

CloudShell

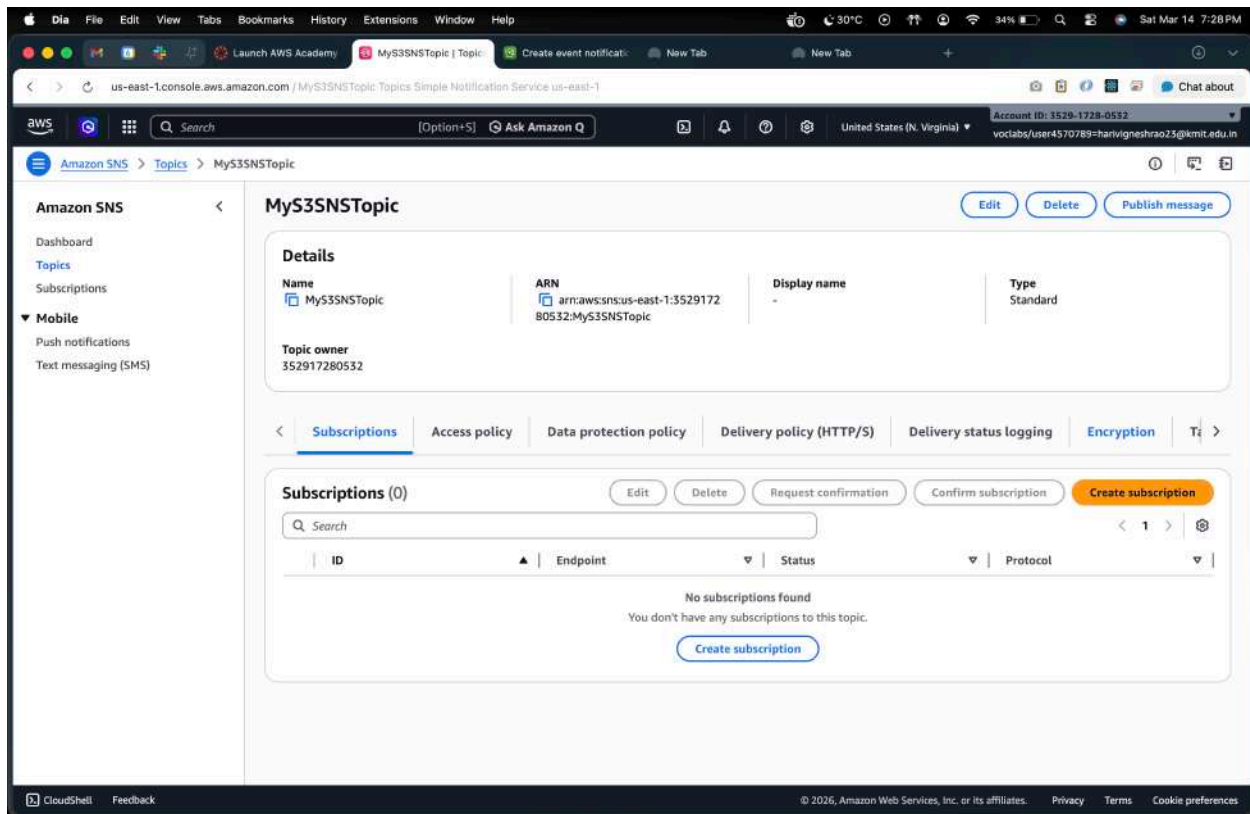
Feedback

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Step 4: Subscribe SQS to SNS

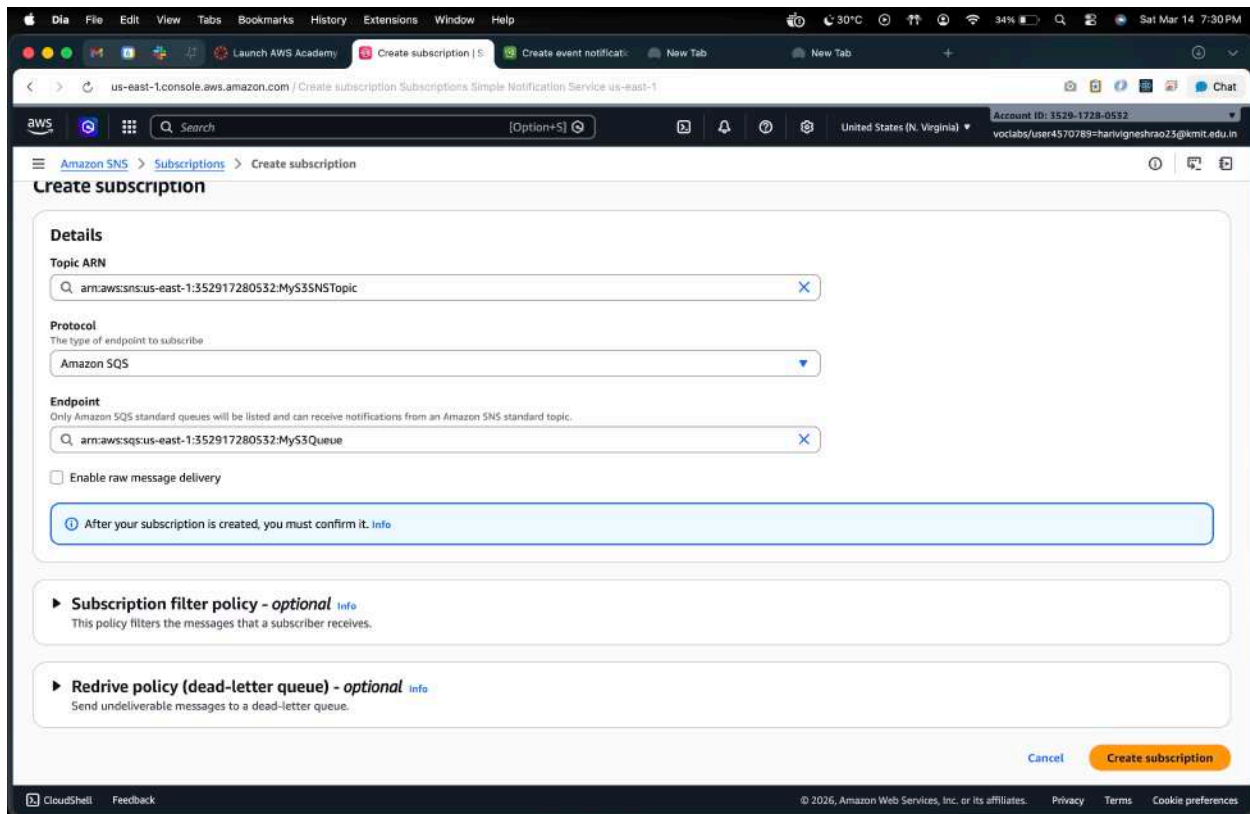
1. Open **MyS3NSTopic**
2. Click **Create subscription**



Configure:

- Protocol → **Amazon SQS**
- Endpoint → Select **MyS3Queue**

3. Click **Create subscription**



Step 5: Add Access Policy to SQS

Open **MyS3Queue** → **Access Policy** → **Edit**

Replace:

- SQS ARN

```
{
```

```
  "Version": "2012-10-17",
```

```
  "Id": "__default_policy_ID",
```

```
  "Statement": [
```

```
    {
```

```
      "Sid": "__owner_statement",
```

```
      "Effect": "Allow",
```

```
      "Principal": {
```

```
"AWS": "arn:aws:iam::352917280532:root"

},

"Action": "SQS:*",

"Resource": "arn:aws:sqs:us-east-1:352917280532:MyS3Queue"

},

{

  "Sid": "Allow-SNS-SendMessage",

  "Effect": "Allow",

  "Principal": {

    "Service": "sns.amazonaws.com"

  },

  "Action": "SQS:SendMessage",

  "Resource": "arn:aws:sqs:us-east-1:352917280532:MyS3Queue",

  "Condition": {

    "ArnEquals": {

      "aws:SourceArn": "arn:aws:sns:us-east-1:352917280532:MyS3SNSTopic"

    }

  }

}

]

} (Modern AWS console often auto-generates this when configuring from S3)
```

Click **Save**.

us-east-1.console.aws.amazon.com / Edit queue Simple Queue Service us-east-1

Amazon SQS > Queues > MySQQueue > Edit

Access policy Info

Define who can access your queue.

```
1 {
2   "Version": "2012-10-17",
3   "Id": "__default_policy_ID",
4   "Statement": [
5     {
6       "Sid": "__owner_statement",
7       "Effect": "Allow",
8       "Principal": {
9         "AWS": "arn:aws:iam::352917280532:root"
10      },
11      "Action": "SQS:*",
12      "Resource": "arn:aws:sqs:us-east-1:352917280532:MySQQueue"
13    },
14    {
15      "Sid": "Allow-SNS-SendMessage",
16      "Effect": "Allow",
```

8:21 JSON

[Policy generator](#)

Redrive allow policy - Optional Info

Identify which source queues can use this queue as the dead-letter queue.

Select which source queues can use this queue as the dead-letter queue.

☒ Disabled

☐ Enabled

us-east-1.console.aws.amazon.com / Edit queue Simple Queue Service us-east-1

Amazon SQS > Queues > MySQQueue > Edit

[Policy generator](#)

Redrive allow policy - Optional Info

Identify which source queues can use this queue as the dead-letter queue.

Select which source queues can use this queue as the dead-letter queue.

☒ Disabled

☐ Enabled

Dead-letter queue - Optional Info

Send undeliverable messages to a dead-letter queue.

Set this queue to receive undeliverable messages.

☒ Disabled

☐ Enabled

Tags - Optional Info

A tag is a label assigned to an AWS resource. Use tags to search and filter your resources or track your AWS costs.

No tags associated with this queue.

[Add new tag](#)

You can add 50 more tags.

[Cancel](#) [Save](#)

Step 6: Add Access Policy to SNS

Open **SNS Topic** → **Edit access policy**

Replace:

- SNS topic ARN
- S3 bucket ARN
- Account ID

```
{  
  
  "Version": "2012-10-17",  
  
  "Id": "__default_policy_ID",  
  
  "Statement": [  
  
    {  
  
      "Sid": "__default_statement_ID",  
  
      "Effect": "Allow",  
  
      "Principal": {  
  
        "AWS": "*"
      },  
  
      "Action": [  
  
        "SNS:Publish",  
  
        "SNS:RemovePermission",  
  
        "SNS:SetTopicAttributes",  
  
        "SNS>DeleteTopic",  
  
        "SNS:ListSubscriptionsByTopic",  
  
        "SNS:GetTopicAttributes",
```

```
"SNS:AddPermission",
  "SNS:Subscribe"
],
"Resource": "arn:aws:sns:us-east-1:352917280532:MyS3SNSTopic",
"Condition": {
  "StringEquals": {
    "AWS:SourceAccount": "352917280532"
  }
},
{
  "Sid": "AllowS3Publish",
  "Effect": "Allow",
  "Principal": {
    "Service": "s3.amazonaws.com"
  },
  "Action": "SNS:Publish",
  "Resource": "arn:aws:sns:us-east-1:352917280532:MyS3SNSTopic",
  "Condition": {
    "ArnLike": {
      "aws:SourceArn": "arn:aws:s3:::mys3eventbucket1234salaar"
    }
  },
  "StringEquals": {
```

"aws:SourceAccount": "352917280532"

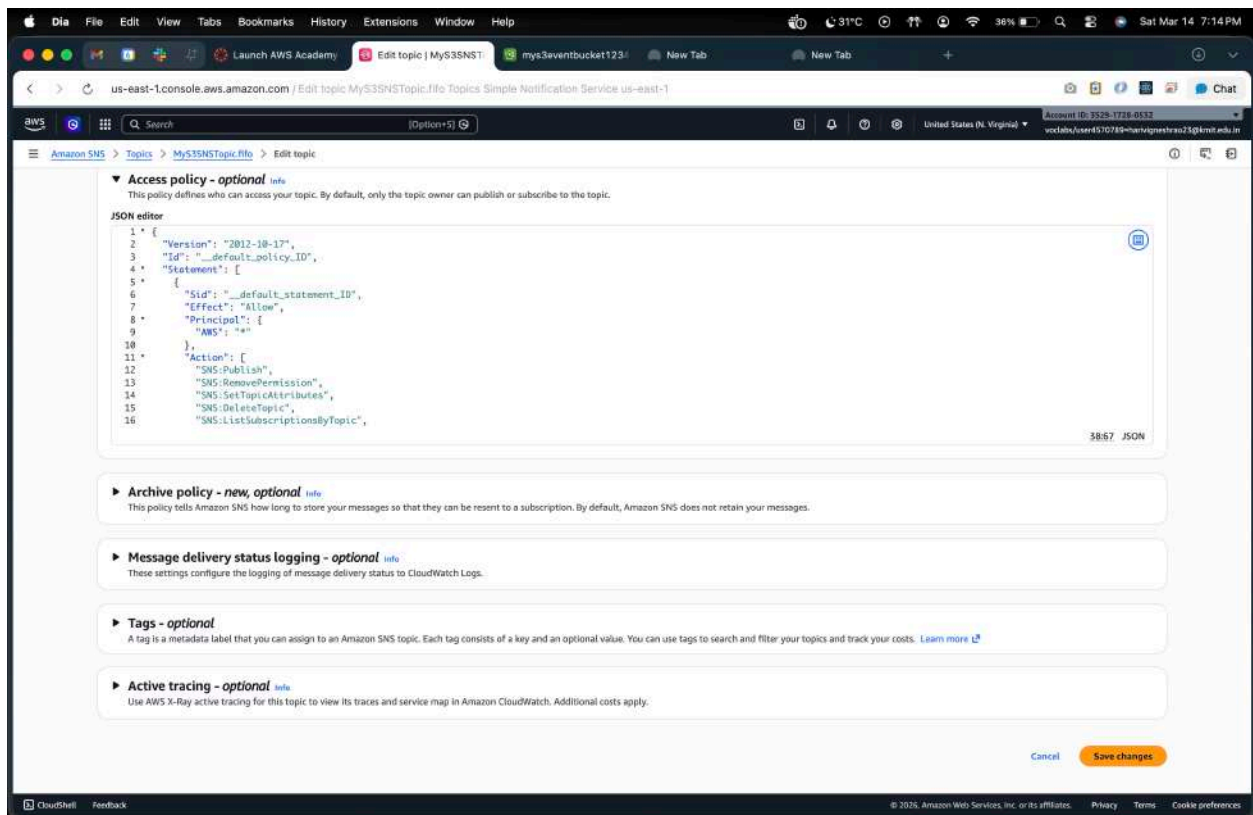
}

}

}

]

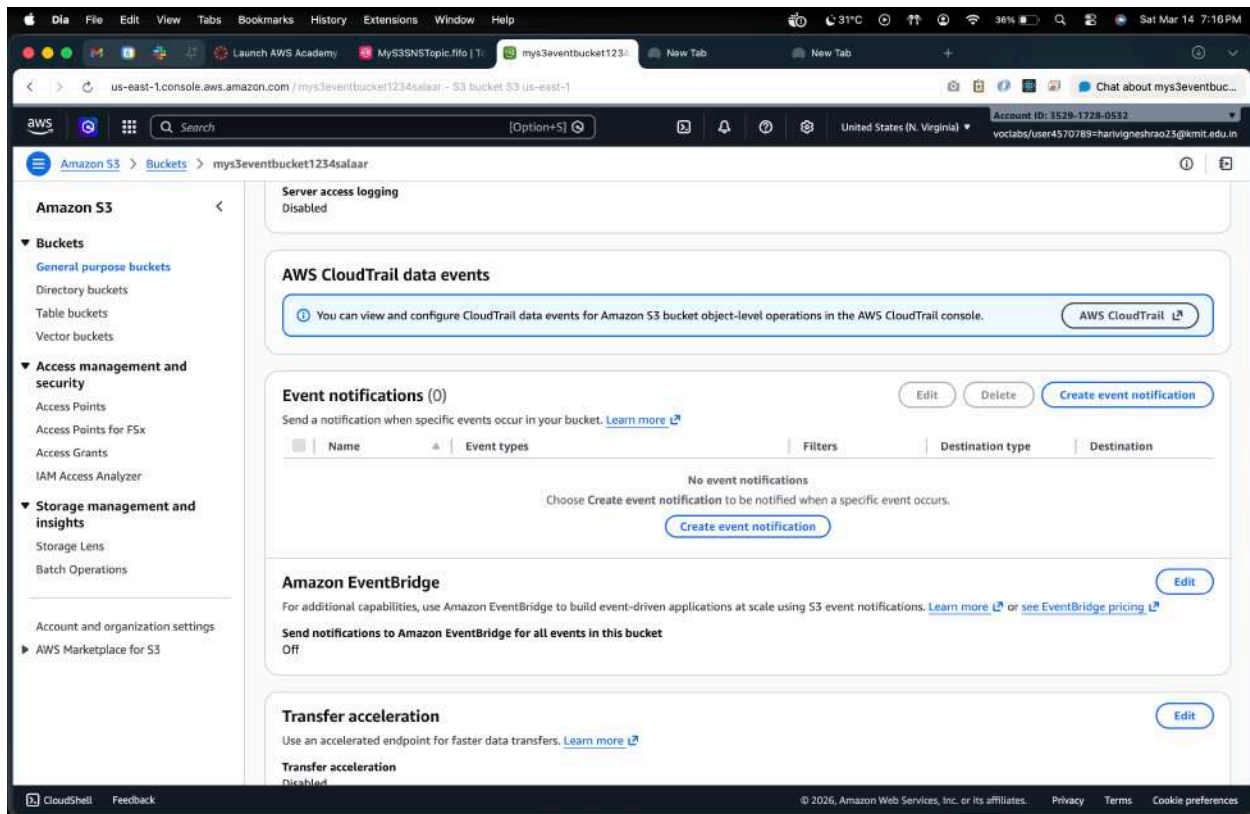
} Save the policy.



Step 7: Configure S3 Event Notification

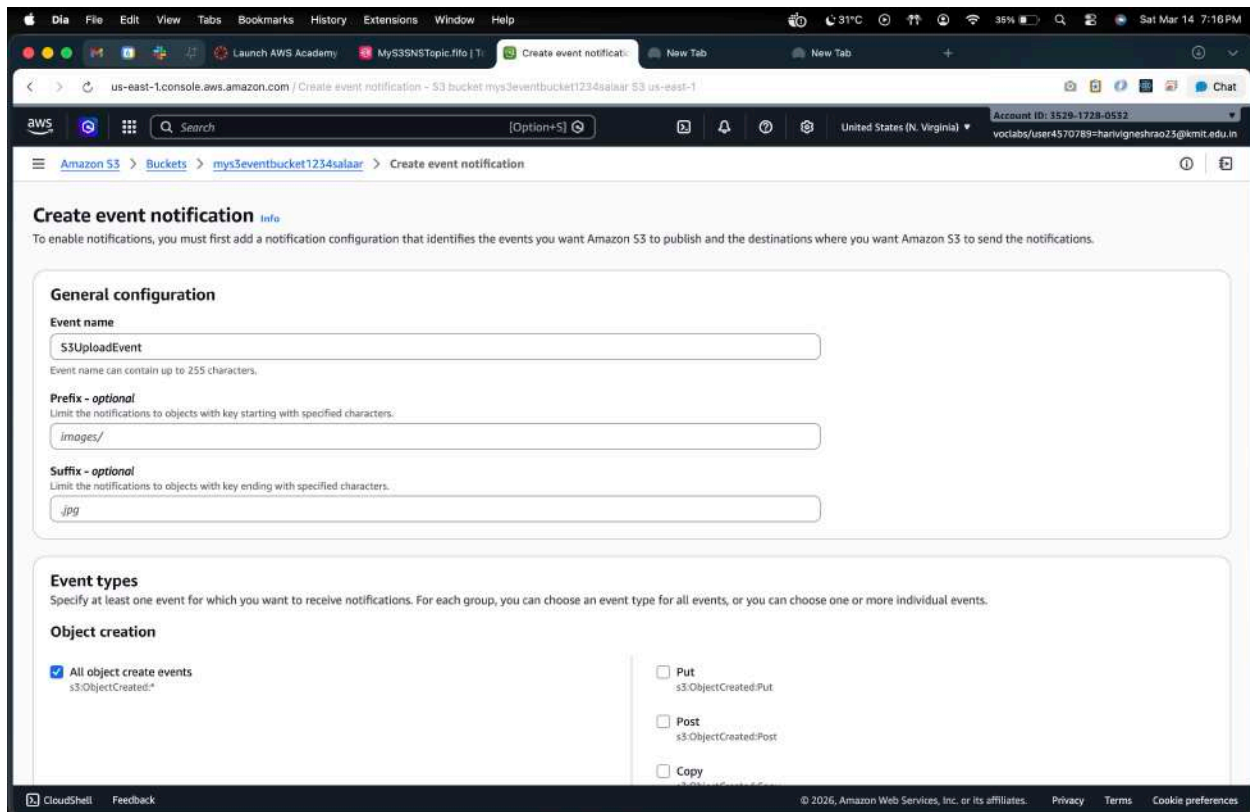
1. Open **S3 Bucket (mys3eventbucket123)**
2. Go to **Properties**
3. Scroll to **Event notifications**

Click **Create event notification**



Configure:

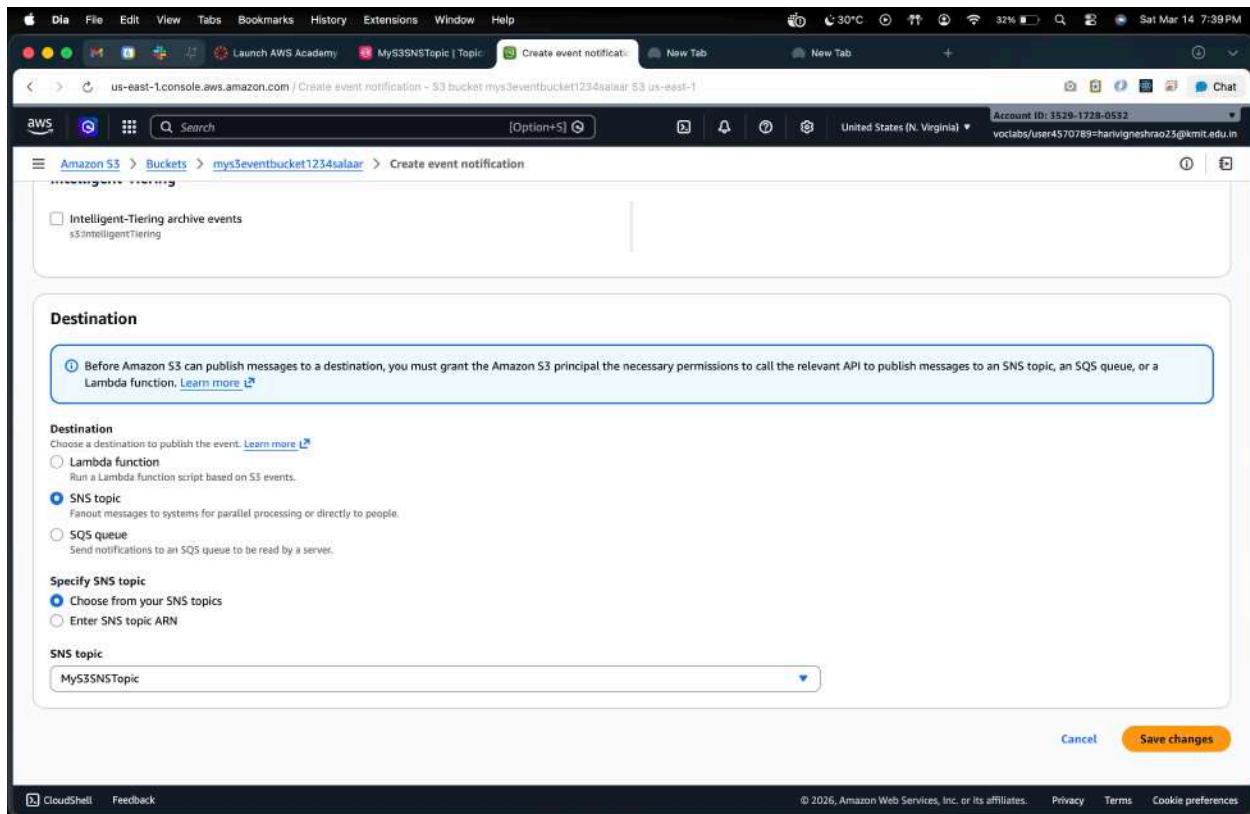
- Event name → **S3UploadEvent**
- Event type → **All object create events**



Destination:

• SNS Topic → MyS3SNSTopic

Click **Save changes**.



Step 8: Test the Architecture

1. Open the **S3 bucket**
2. Click **Upload file**
3. Upload any file (example: **test.txt**)

Now:

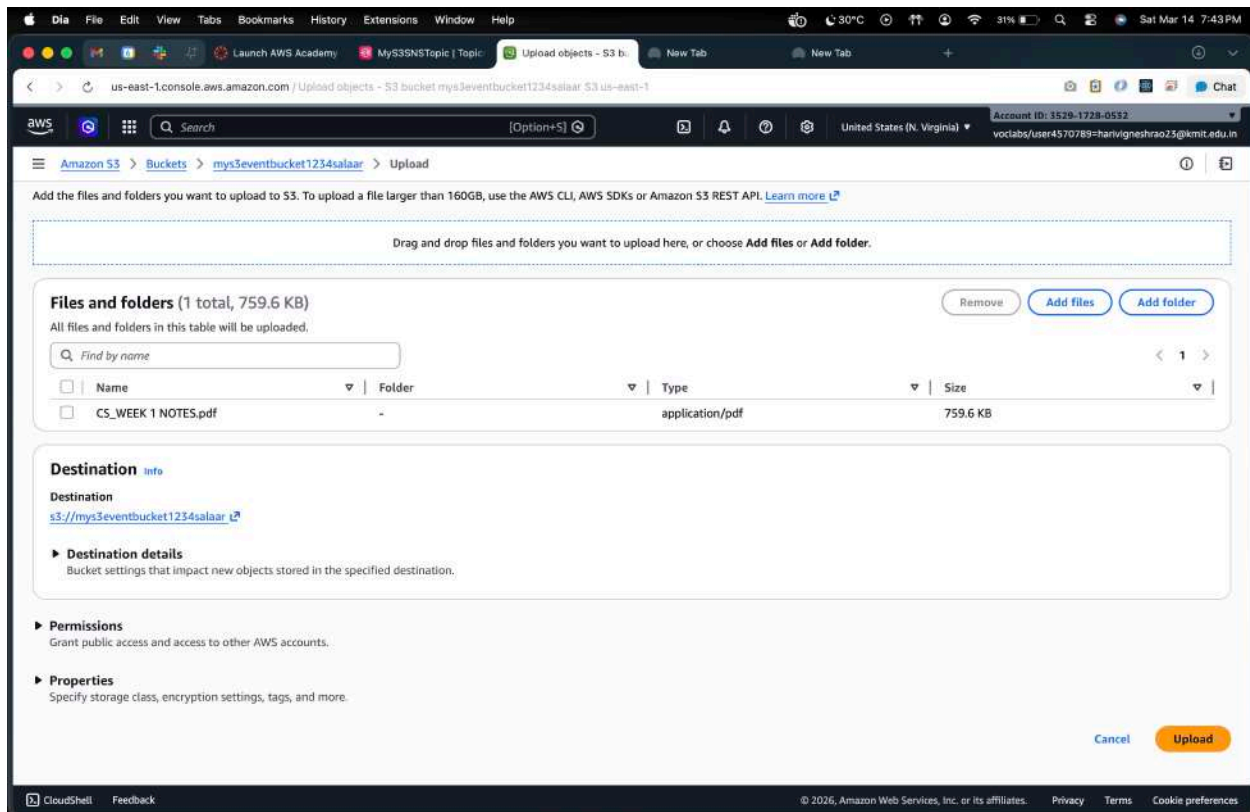
File Upload → S3



Notification → SNS



Message → SQS



Step 9: Verify Message in SQS

1. Open **MyS3Queue**
2. Click **Send and receive messages**
3. Click **Poll for messages**

You will see the **S3 event notification message**.

us-east-1.console.aws.amazon.com / Send and receive messages Simple Queue Service us-east-1

Message group ID - optional, new [Info](#)

A group identifier for the message to allow fair processing across message groups in a standard queue.

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation ("'%&()*+,-./:;<=>@[\]^_`{|}~").

Delivery delay [Info](#)

The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.

0 Seconds

Should be between 0 seconds and 15 minutes.

► Message attributes - optional [Info](#)

Clear content Send message

Receive messages [Info](#)

Edit poll settings Stop polling Poll for messages

Messages available 2 Polling duration 30 Maximum message count 10 Polling progress 67% 1 receives/second

Messages (2)

Search messages

☐	ID	Sent	Size	Receive count
☐	652a7560-0fd9-4de9-93dc-039789a5a6b2	2026-03-14T19:40+05:30	1.18 KB	1
☐	e9756834-a119-440b-9148-ff336d4017c9	2026-03-14T19:43+05:30	1.86 KB	1

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us-east-1.console.aws.amazon.com / Send and receive messages Simple Queue Service us-east-1

Message group ID - optional, new [Info](#)

A group identifier for the message to allow fair processing across message groups in a standard queue.

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation ("'%&()*+,-./:;<=>@[\]^_`{|}~").

Delivery delay [Info](#)

The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.

0 Seconds

Should be between 0 seconds and 15 minutes.

► Message attributes - optional [Info](#)

Clear content Send message

Receive messages [Info](#)

Edit poll settings Stop polling Poll for messages

Messages available 2 Polling duration 30 Maximum message count 10 Polling progress 67% 1 receives/second

Messages (2)

Search messages

☐	ID	Sent	Size	Receive count
☐	652a7560-0fd9-4de9-93dc-039789a5a6b2	2026-03-14T19:40+05:30	1.18 KB	1
☐	e9756834-a119-440b-9148-ff336d4017c9	2026-03-14T19:43+05:30	1.86 KB	2

Received message: 652a7560-0fd9-4de9-93dc-039789a5a6b2

Body Attributes Details

Subject: Amazon S3 Notification,
Message: {"Service":"Amazon S3","Event":"s3:TestEvent","Time":"2026-03-14T14:10:03.940Z","Bucket":"mys3eventbucket1234salaar","Requestid":"X1Q7FRB16GHWD8GT","Hostid":"X1Q7FRB16GHWD8GT"}

Done

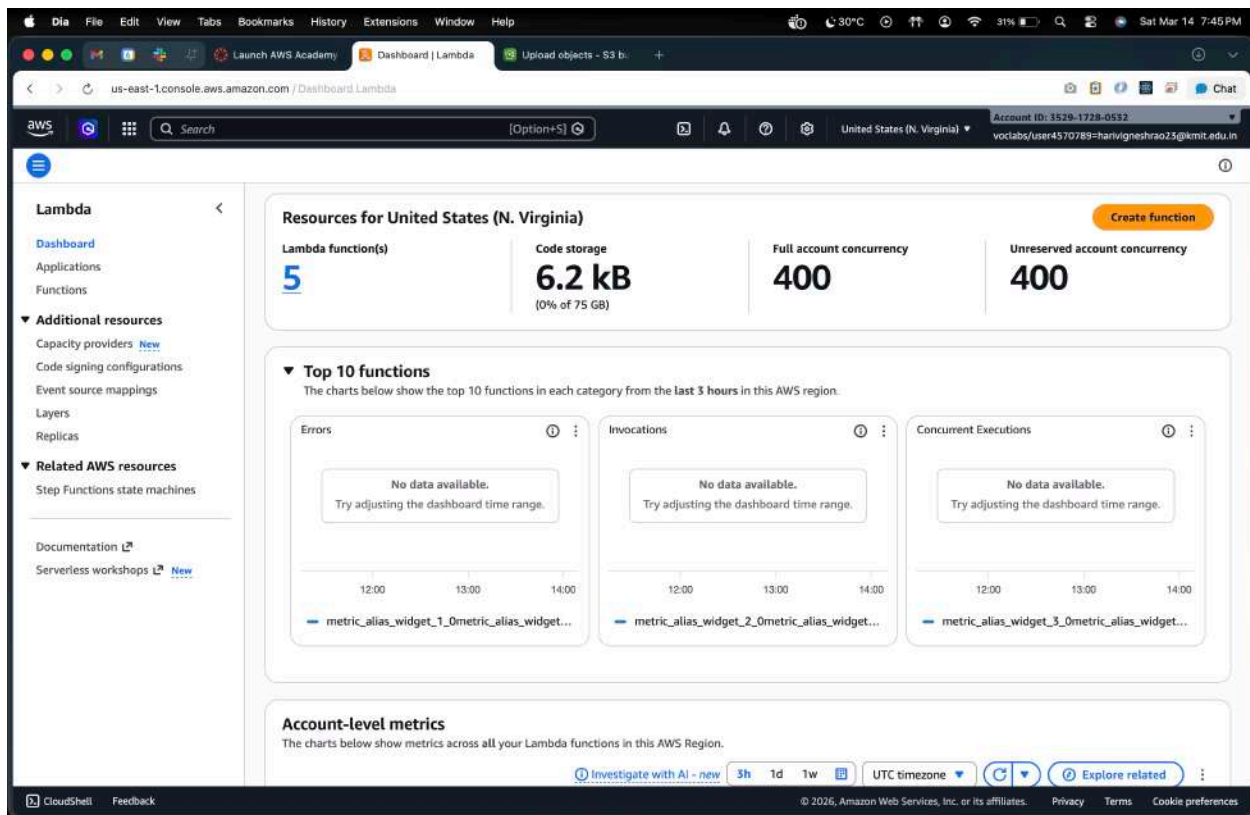
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Step 10: Configure Lambda as Consumer

Now configure Lambda to **automatically process messages from SQS**.

Create Lambda Function

1. Search **Lambda** in AWS Console
2. Open **AWS Lambda**
3. Click **Create Function**



Configure:

- Author from scratch
- Function name → **SQSConsumerFunction**
- Runtime → **Python 3.x**

Click **Create function**

FileEditViewTabsBookmarksHistoryExtensionsWindowHelp

Launch AWS AcademyCreate function | Func

us-east-1.console.aws.amazon.com / Create function Functions Lambda

aws

Search[Option+S]

United States (N. Virginia)

Account ID: 3529-1728-0532voclabs/user4570789-harivigneshrao23@kmit.edu.in

Lambda > Functions > Create function

1

Create function [Info](#)

Choose one of the following options to create your function.

☒ Author from scratch
Start with a simple Hello World example.

☐ Use a blueprint
Build a Lambda application from sample code and configuration presets for common use cases.

☐ Container image
Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

SQSConsumerFunction

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime

[Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

Durable execution - new

[Info](#)

Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).

☐ Enable

Architecture

[Info](#)

Choose the instruction set architecture you want for your function code.

☐ arm64

☒ x86_64

Permissions

[Info](#)

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us-east-1.console.aws.amazon.com / Create function Functions Lambda

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Search[Option+S]

United States (N. Virginia)

Account ID: 3529-1728-0532voclabs/user4570789-harivigneshrao23@kmit.edu.in

Lambda > Functions > Create function

1

Durable execution - new

[Info](#)

Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).

☐ Enable

Architecture

[Info](#)

Choose the instruction set architecture you want for your function code.

☐ arm64

☒ x86_64

Permissions

[Info](#)

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role

☒ Use another role

LabRole

[Create new role](#)

[View role details in IAM](#)

► Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

Cancel

Create function

CloudShellFeedback

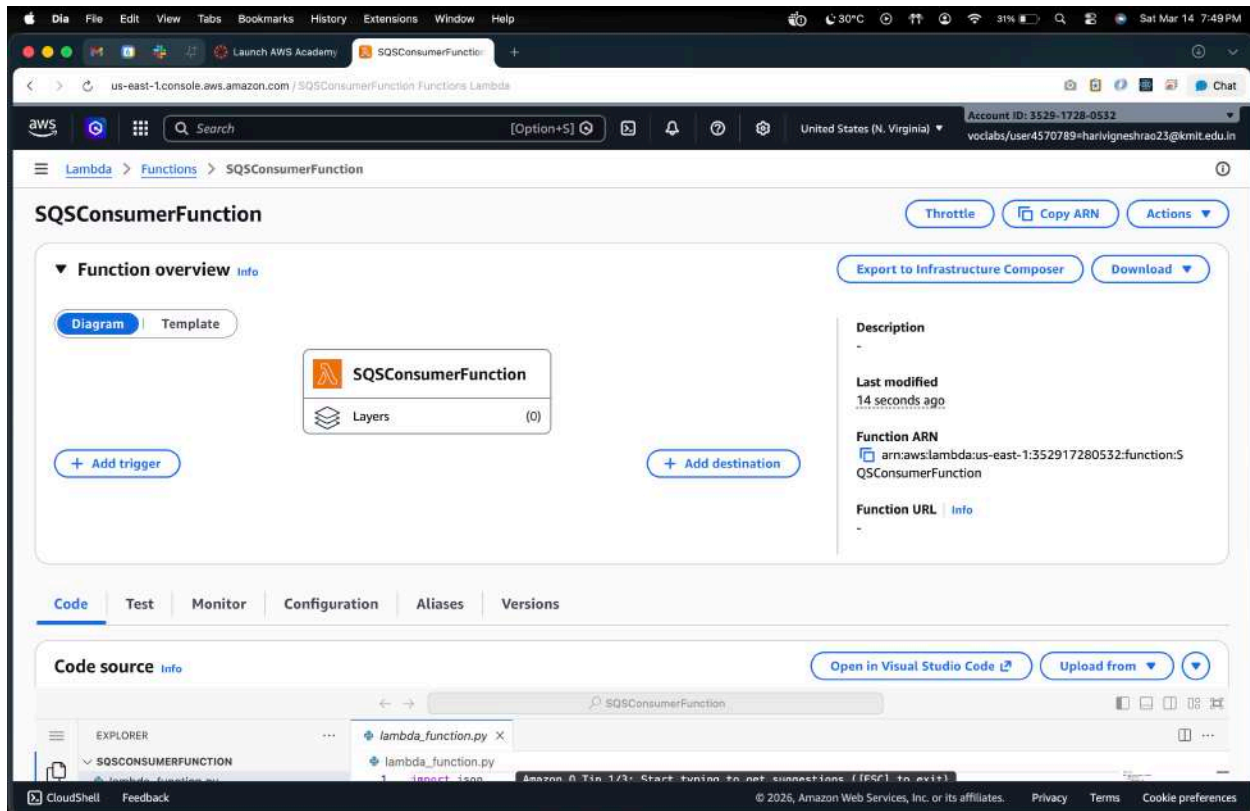
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Add SQS Trigger

1. Open **SQSConsumerFunction**
2. Click **Add Trigger**
3. Select **SQS**
4. Choose **MyS3Queue**

Click **Add**

Now Lambda will automatically read messages from SQS.



Dia

File

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30°C

30%

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Add triggers | Lambda

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aws

Search

[Option+S]

United States (N. Virginia)

Account ID: 3529-1728-0532

voclabs/user4570789-harivigneshrao23@kmit.edu.in

Lambda

Add triggers

Add trigger

Trigger configuration

Info

SQS

aws event-source-mapping polling queue

SQS queue

Choose or enter the ARN of an SQS queue.

arn:aws:sqs:us-east-1:352917280532:MyS3Queue

X

Event poller configuration

☒ Activate trigger

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

Provisioned mode

Info

Select to configure provisioned mode for your event source mapping, which allows you to optimize the throughput of your event-driven applications.

☐ Configure

☐ Enable EventCount metrics

Track the number of events polled, filtered, invoked, and dropped by your event source mapping. CloudWatch charges apply.

Batch size - optional

The maximum number of records in each batch to send to the function.

10

The maximum is 10,000 for standard queues and 10 for FIFO queues.

Batch window - optional

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voclabs/user4570789-harivigneshrao23@kmit.edu.in

Lambda

Add triggers

0

When the batch size is greater than 10, set the batch window to at least 1 second.

Maximum concurrency - optional

The maximum number of concurrent function instances that the SQS event source can invoke.

Specify a value between 2 and 1000. To deactivate, leave the box empty.

Report batch item failures - optional

Allow your function to return a partial successful response for a batch of records.

☐

Filter criteria - optional

Define the filtering criteria to determine whether or not to process an event. Each filter must be in a valid JSON format in filter rule syntax. [Learn more](#)

Add

During trigger creation, Lambda translates your filter(s) into a single JSON structure containing all your filtering criteria.

Info

☐ Encrypt filter criteria with customer managed KMS key

Lambda will use this key to encrypt and decrypt your resources.

► Additional settings

In order to read from the SQS trigger, your execution role must have proper permissions.

Cancel

Add

CloudShell

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Lambda Code

Replace code with:

```
def lambda_handler(event, context):

    for record in event['Records']:

        print("Message received from SQS:")

        print(record['body'])

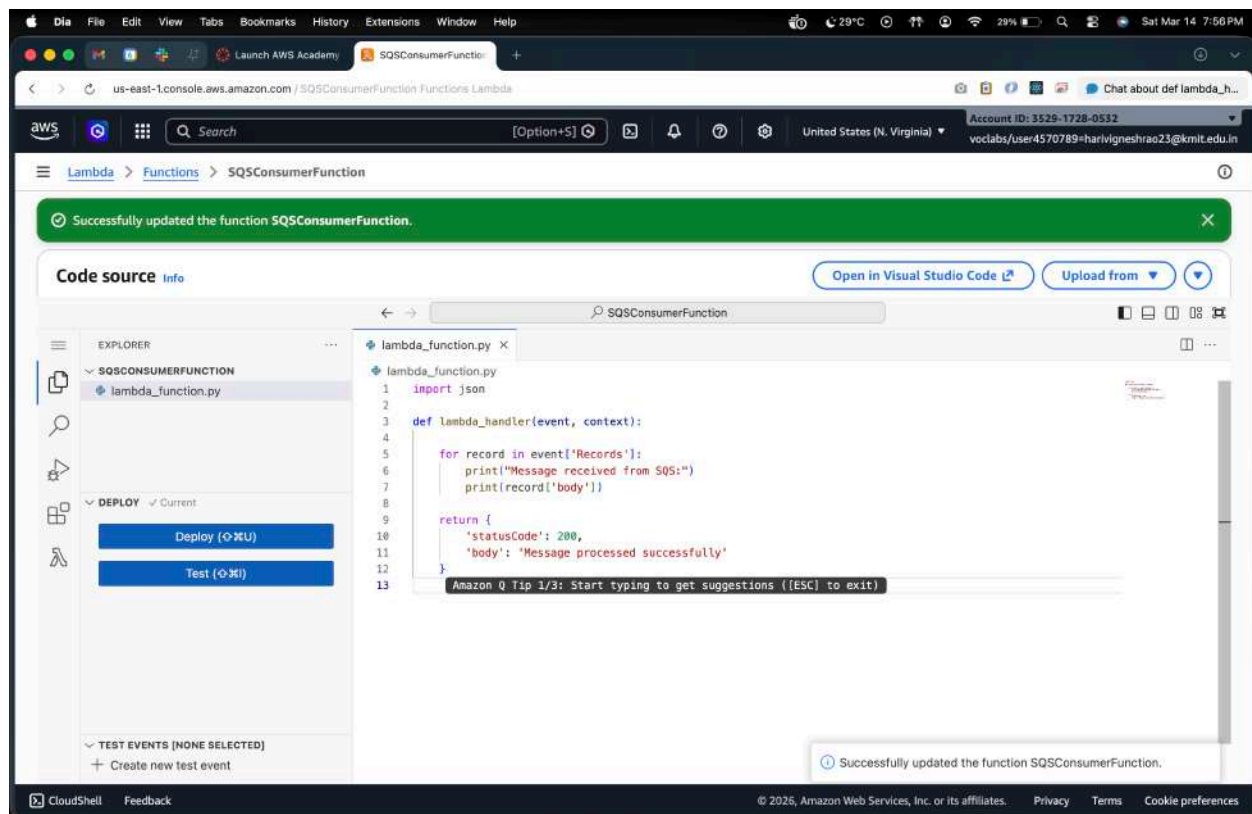
    return {

        'statusCode': 200,

        'body': 'Message processed successfully'

    }
```

Click **Deploy**.



Final Architecture

User uploads file



Amazon S3



SNS Topic



SQS Queue



Lambda Function (Consumer)



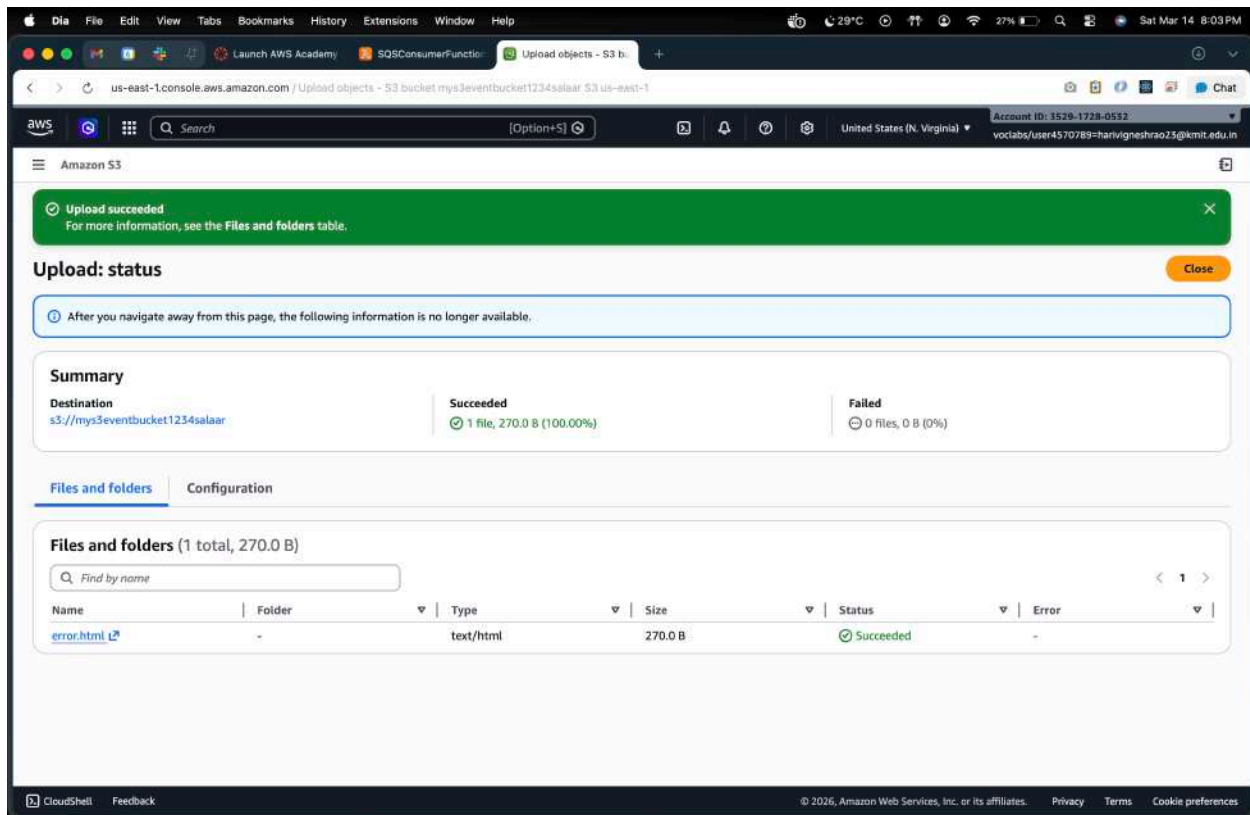
Processing / Logs / Database

Result

When a file is uploaded to S3:

1. S3 sends a notification to SNS
2. SNS forwards the message to SQS
3. SQS stores the message
4. Lambda automatically reads and processes the message

This creates a **serverless event-driven architecture commonly used in real cloud applications.**



us-east-1.console.aws.amazon.com / Send and receive messages Simple Queue Service us-east-1

Amazon SQS > Queues > MySQQueue > Send and receive messages

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (!"#\$%&'()*+,-./:;<=>?@[\]^_`{|}~).

Delivery delay [Info](#)
The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.
 Seconds
Should be between 0 seconds and 15 minutes.

► **Message attributes** - optional [Info](#)

[Clear content](#) [Send message](#)

Receive messages [Info](#) [Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

Messages available 0 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 3%
0 receives/second

Messages (0) [View details](#) [Delete](#)

☐ ID ☐ Sent ☐ Size ☐ Receive count

No messages. To view messages in the queue, poll for messages.

[Poll for messages](#)

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us-east-1.console.aws.amazon.com / CloudWatch us-east-1

CloudWatch > Log management > /aws/lambda/SQSConsumerFunction > 2026/03/14/[SLATEST]62c007ebc2584eef915a04af605534e

CloudWatch [Log Management](#)

Log events [Actions](#) [Start tailing](#) [Create metric filter](#)

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

[Clear](#) [1m](#) [30m](#) [1h](#) [12h](#) [Custom](#) [UTC timezone](#)

[Display](#)

Timestamp	Message
	No older events at this moment. Retry
2026-03-14T14:33:46.724Z	INIT_START Runtime Version: python:3.14.v35 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:6e4c2a8804e47c3599..
2026-03-14T14:33:46.878Z	START RequestId: 009d8ca3-9727-5a36-98b9-ee8ce4dc78b Version: SLATEST
2026-03-14T14:33:46.878Z	Message received from SQS:
2026-03-14T14:33:46.878Z	[
2026-03-14T14:33:46.878Z	"Type": "Notification",
2026-03-14T14:33:46.878Z	"MessageId": "3b9f9444-7dc8-54f4-9b64-f66b09c2ed6c",
2026-03-14T14:33:46.878Z	"TopicArn": "arn:aws:sns:us-east-1:352917280532:MyS3SNSTopic",
2026-03-14T14:33:46.878Z	"Subject": "Amazon S3 Notification",
2026-03-14T14:33:46.878Z	"Message": "[{"Records":[{"eventVersion":"2.1","eventSource":"aws:s3","awsRegion":"us-east-1","eventTi...
2026-03-14T14:33:46.878Z	"Timestamp": "2026-03-14T14:33:46.416Z",
2026-03-14T14:33:46.878Z	"SignatureVersion": "1",
2026-03-14T14:33:46.878Z	"Signature": "lWHEcNH0+cKZv9xFAJE2bg6YcmZ+6N20jHbInqtVGFLarnOr7IA0GqxvgTwG89HlHeSgn8sZEVXbbfXX43y5fwupl1kIrcGW..
2026-03-14T14:33:46.878Z	"SigningCertURL": "https://sns.us-east-1.amazonaws.com/SimpleNotificationService-75061e35b36eF5a444dd1a8e7cc3ed8.pe...
2026-03-14T14:33:46.878Z	"UnsubscribeURL": "https://sns.us-east-1.amazonaws.com/?Action=Unsubscribe&SubscriptionArn=arn:aws:sns:us-east-1:352...
2026-03-14T14:33:46.878Z	}

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